

2018

ممثل

قسم هندسة الطاقة الكهربائية

مصطفى القريشي

سنوات دوائر كهربائية ١

سيركت 1

امتحانات الفاينل

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طلاب البور والإتصالات

spark team

spark-fet.com

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هند خليل حجي سليمان

Al-Balqa' Applied University
Faculty of
EngineeringTechnology
Electrical circuits 1

جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجيه

Final exam 4-08-2015

Q1: Find the voltage (v) for the network shown in Fig 1 using superposition theorem. (10marks)

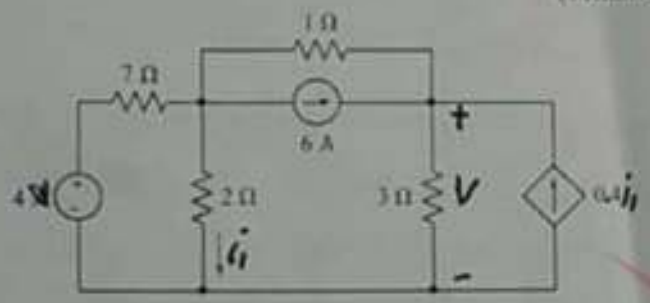


Fig 1

Q2: For the circuit shown in Fig 2: (10marks)
a) Determine the value of Rl to which a maximum power can be delivered .
b) The maximum power deliverd to Rl.

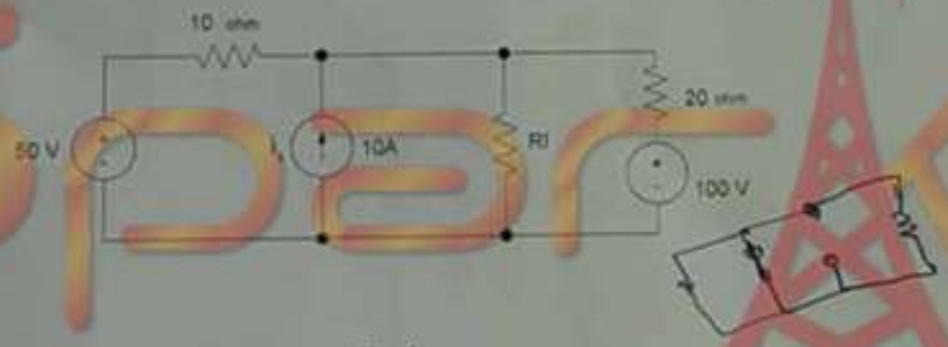


Fig 2

Q3: a) For the circuit shown in Fig 3.a find the equivalent capacitance (10 marks)

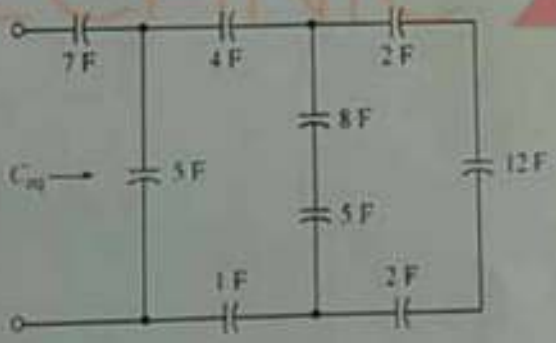


Fig 3.a

b) Calculate the power dissipated in the 40 Ω resistor and the voltage labeled Vc for the circuit given in fig 3.b

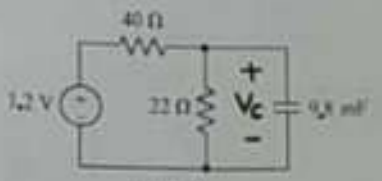
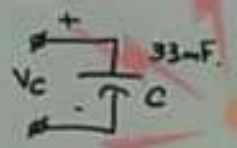


fig 3.b

Q4: The voltage across a 33 mF capacitor is shown graphically in Fig 4 : (8 marks)

a) Assuming the passive sign convention, sketch the resulting current waveform across the capacitor.

b) Find the energy stored in the capacitor at t=20ms.



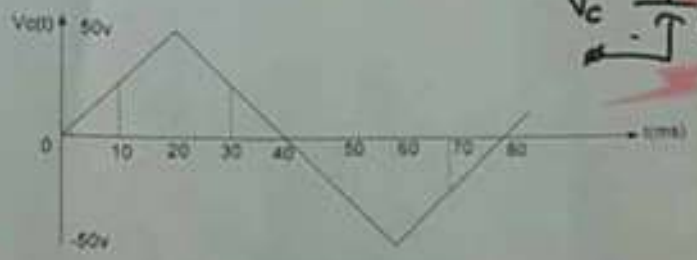


Fig 4

Q5: For the circuit shown in Fig 5 the switch opens at t=0 after having been closed for a long time. Find: (12 marks)

a) $i_1(t)$ for $t >= 0^+$.
b) $i_L(t)$ for $t >= 0^+$.
c) $V_2(t)$ for $t >= 0^+$.

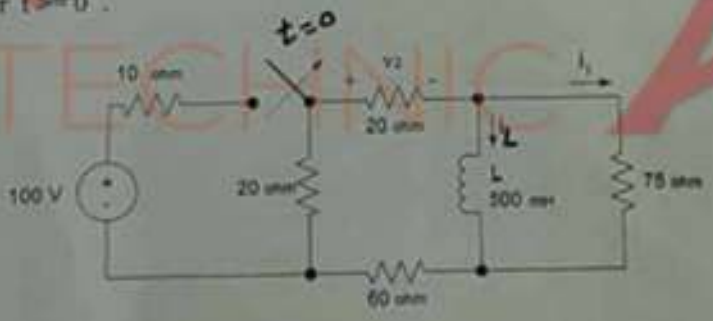
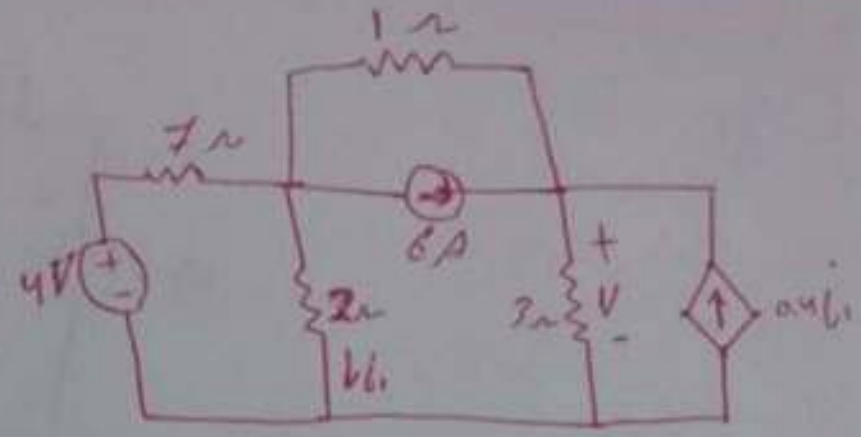


Fig 5

Good Luck

Q1

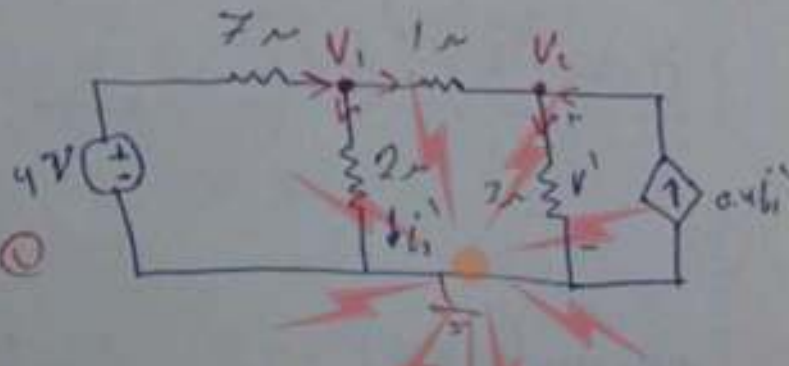


$V = V' + V''$

$= 0.69 + 2.682$

$V = 3.372$ Volt

at node (V_1)



$\frac{4 - V_1}{7} = \frac{V_1}{2} + V_1 - V_2$

$\rightarrow \boxed{23V_1 - 14V_2 = 8}$ — (1)

at node (V_2)

$V_1 - V_2 + 0.4\frac{V_1}{2} = \frac{V_2}{3} \rightarrow \boxed{7.2V_1 - 8V_2 = 0}$ — (2)

$V' = V_2 = 0.69$ Volt

$V_1 = 0.76$ Volt

at mesh (1)

$I_1 = 0.4i_1' = 0.4(I_2 - I_3)$

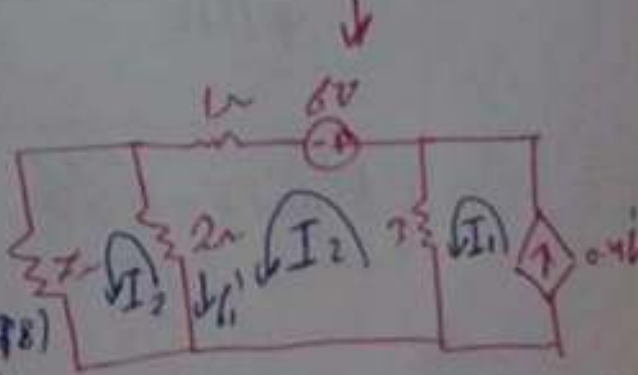


at mesh (2)

$6I_2 - 2I_3 - 3I_1 = -6$

$6I_2 - 2I_3 - 3(0.4(I_2 - I_3)) = -6$

$\boxed{4.8I_2 - 0.8I_3 = -6}$



at mesh (3)

$\boxed{-2I_2 + 9I_1 = 0}$

$I_2 = -1.298$ A

$I_3 = -0.288$ A

$V'' = 7(I_1 - I_2)$

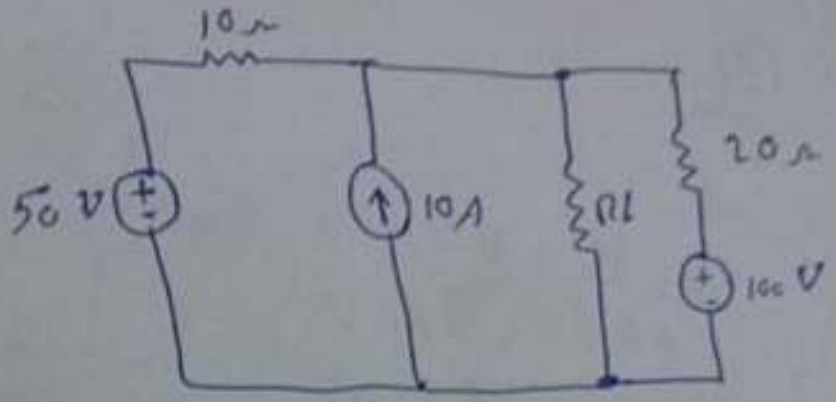
$V'' = 7(-0.404 + 1.298)$

$V'' = 2.682$ Volt

$I_1 = 0.4(-1.298 + 0.288)$

$I_1 = -0.404$

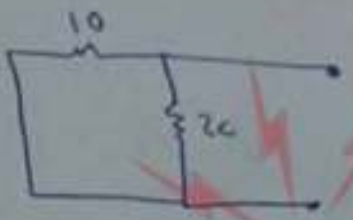
Q2



(a)

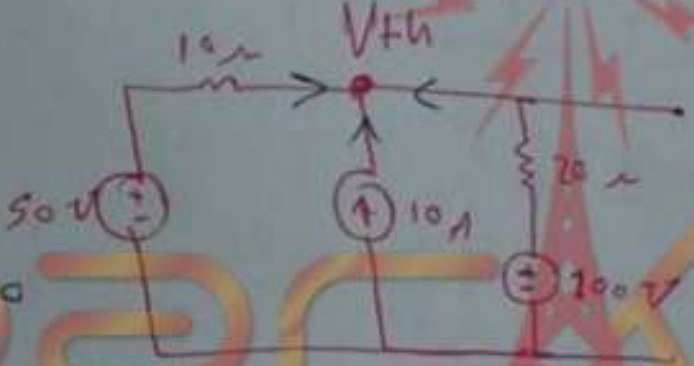
$R_{th} = R_L = 6.667\ \Omega$

←



(b)

at node (V_{th})



$$\frac{50 - V_{th}}{10} + 10 + \frac{100 - V_{th}}{20} = 0$$

$$100 - 2V_{th} + 200 + 100 - V_{th} = 0$$

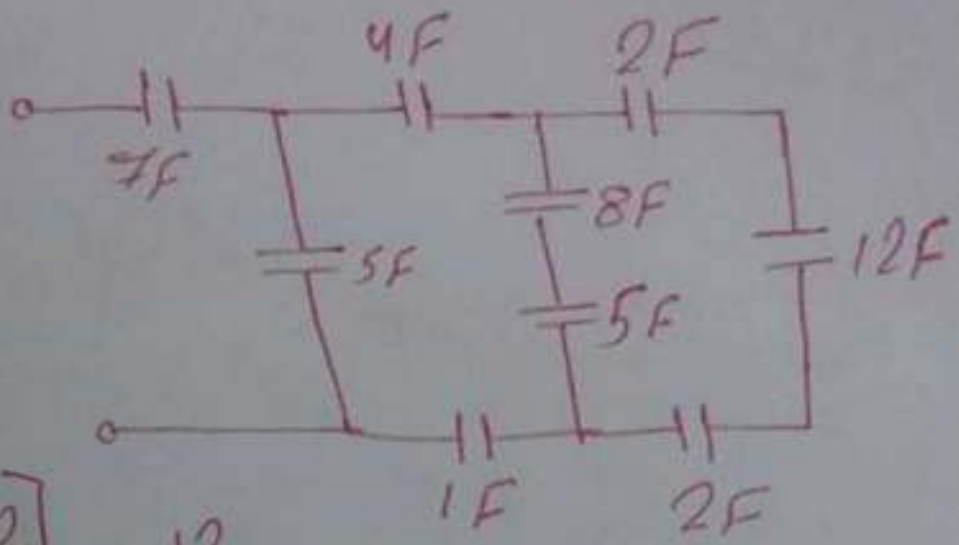
$$3V_{th} = 400 \rightarrow V_{th} = 133.34\ Volts$$

$$P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{(133.34)^2}{4 \times 6.667}$$

~~1667.7 watts~~

$$P_{max} = 666.7\ watts$$

Q3 (a)



[2 series 12 series 2] → $\frac{12}{13}$

[8 series 5] → $\frac{40}{13}$

$\left[\frac{12}{13} \;//\; \frac{40}{13}\right]$ → (4)

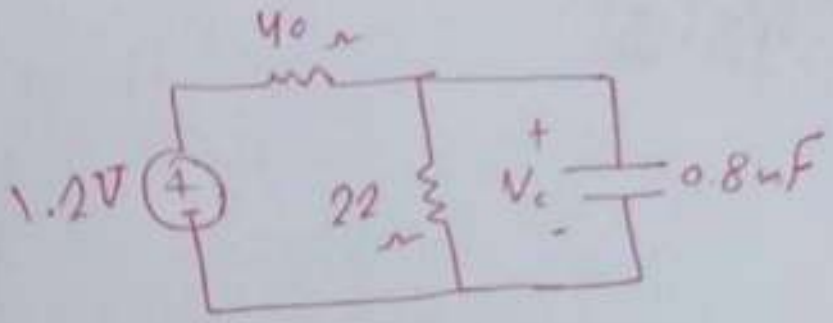
[4 series 4 series 1] → $\frac{2}{3}$

$\left[\frac{2}{3} // 5\right]$ → $\frac{17}{3}$

Ceq ⇒ $\left[\frac{17}{3} \text{ series } 7\right]$

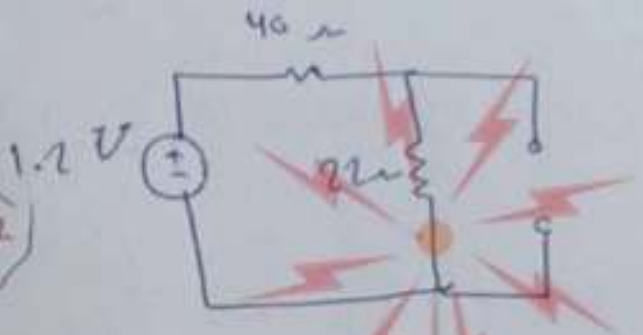
Ceq = 3.13 F

Q3 (b)



$$V_{40\Omega} = \frac{1.2 \times 40}{40 + 22}$$

$$= \frac{24}{31} \text{ Volts}$$



$$P_{40} = \frac{V^2}{R} = 0.0149 \text{ watt}$$

$$V_c = V_{22\Omega} = \frac{1.2 \times 22}{22 + 40} = 0.425 \text{ Volt}$$

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Q4 (a)

$$\text{slope}① = \frac{50-0}{20-0} = 2.5$$

$$y - y_0 = S_0(x - x_0)$$

$$y - 50 = 2.5(x - 20) \rightarrow \boxed{y = 2.5x}$$

$$\text{slope}② = \frac{-50 - 50}{60 - 20} = -2.5$$

$$y + 50 = -2.5(x - 60) \rightarrow \boxed{y = 100 - 2.5x}$$

$$\text{slope}③ = \frac{0 + 50}{80 - 60} = 2.5$$

$$y - 0 = 2.5(x - 80) \rightarrow \boxed{y = 2.5x - 200}$$

$$v(t) = \begin{cases} 2.5t & , 0 < t < 20 \\ 100 - 2.5t & , 20 < t < 60 \\ 2.5t - 200 & , 60 < t < 80 \end{cases}$$

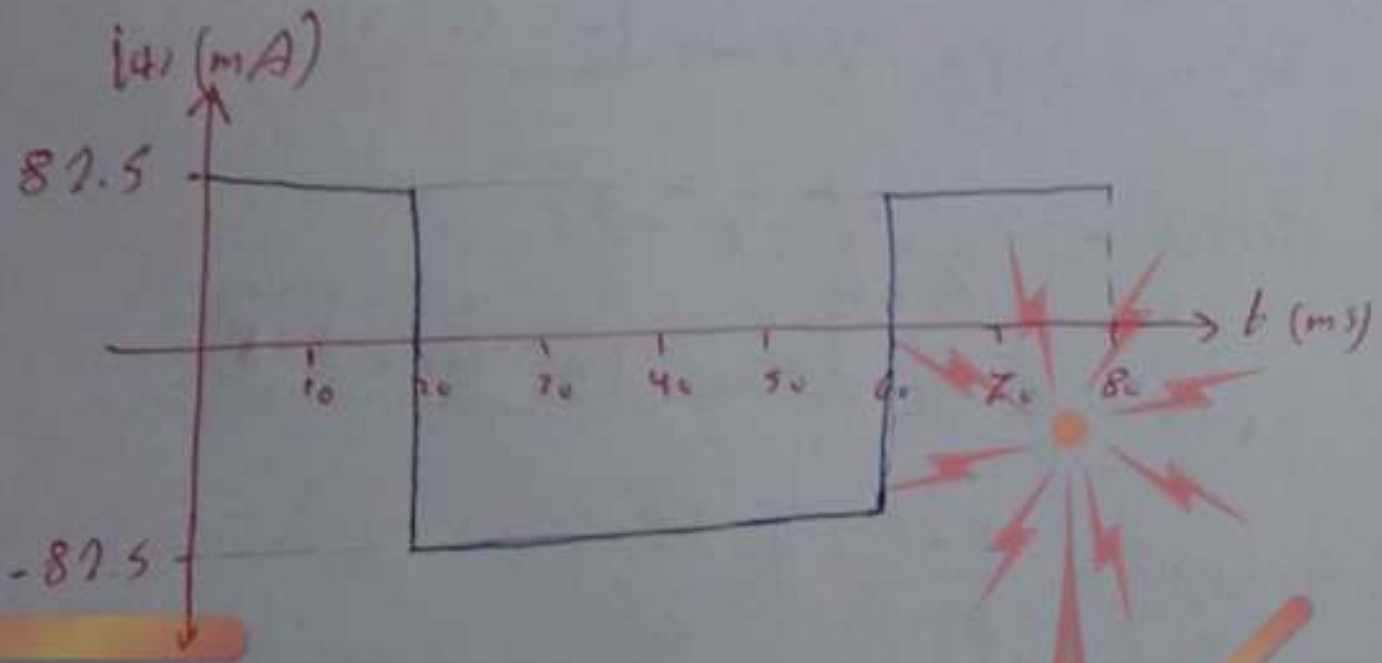
$$i(t) = C\frac{dv}{dt}$$

$$i_1(t) = 33m\,(2.5) = \boxed{82.5\ mA}$$

$$i_2(t) = 33m\,(-2.5) = \boxed{-82.5\ mA}$$

$$i_3(t) = 33m\,(2.5) = \boxed{82.5\ mA}$$

Q4 $i(t) = \begin{cases} 82.5\ mA, & 0 < t < 20 \\ -82.5\ mA, & 20 < t < 60 \\ 82.5\ mA, & 60 < t < 80 \end{cases}$



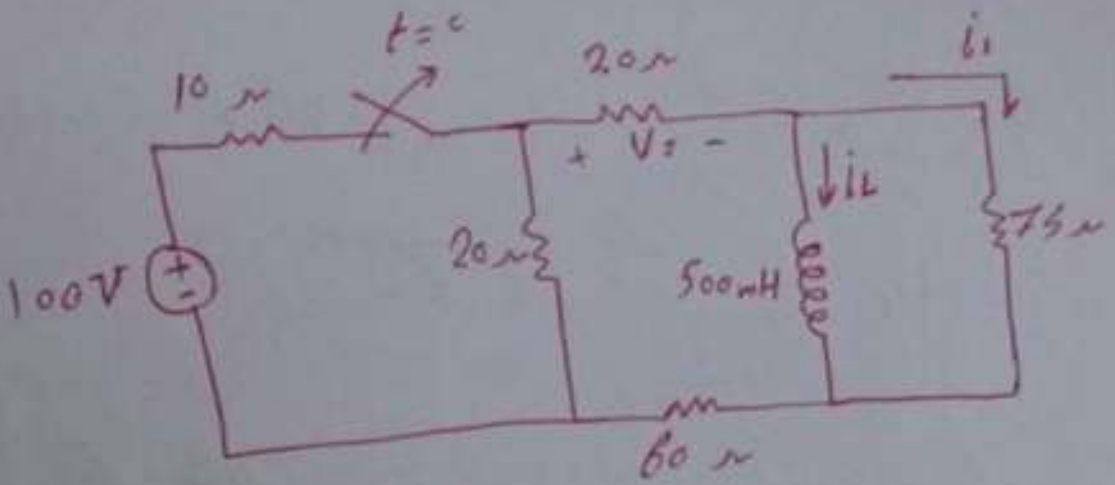
(a) (b) $W = \frac{1}{2} V^2$

$W = \frac{1}{2}(33m)(50)^2$

$W = 41.25\ J$

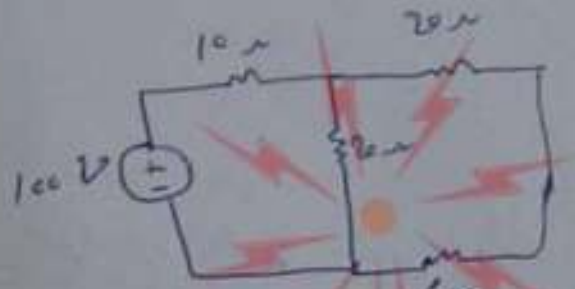
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Q5



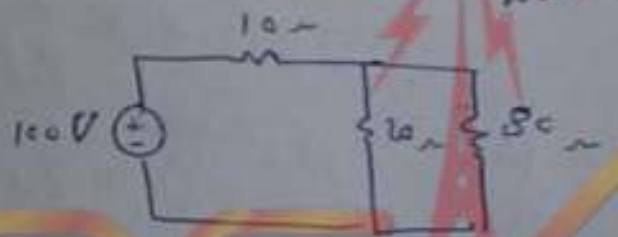
~~(a)~~

$$V_{16\Omega} = \frac{100 \times 16}{16+10} = 61.53\,V_{16}$$

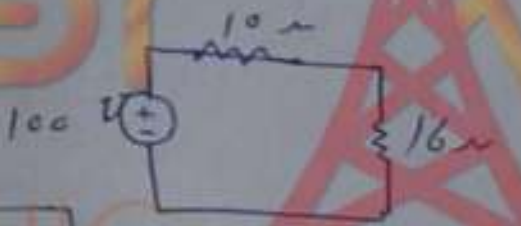


$$I_{L_0} = \frac{I}{8\Omega} = \frac{V}{R} = \frac{61.53}{80}$$

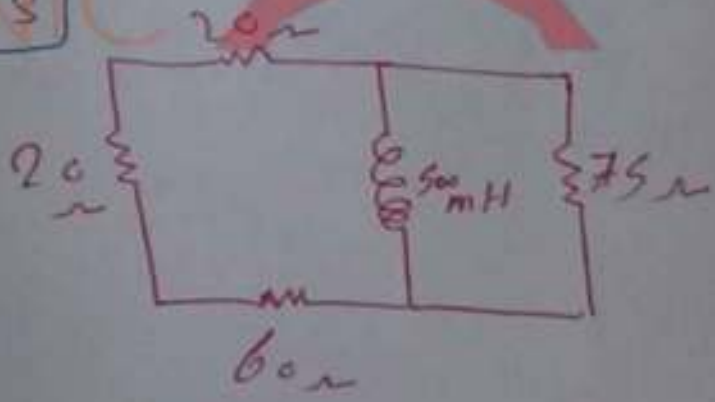
$$I_{L_0} = 0.769\,A$$



$$R_{th} = 42.85\,\Omega$$



$$\tau = \frac{L}{R_{th}} = \frac{500m}{42.85} = 0.0116\,S$$



$$I_L = I_{L(0)}\, e^{-t/\tau}$$

$$I_L = I_{L(0)}\, e^{-t/0.0116}$$

$$I_L = 0.769\, e^{-85.71t}$$

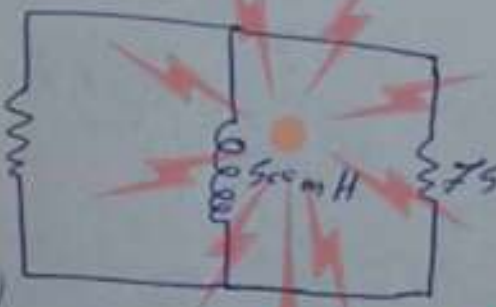
$$I_L = 0.769\, e^{-85.71t}$$

Q5

$$V_L = L\frac{di}{dt} = 0.5\,(0.769 + -85.71\,e^{-85.71t})$$

$$V_L = V_{75\Omega} = -33e^{-85.71t}\ Volt$$

$$I_{75\Omega} = \frac{V}{R} = \frac{-33e^{-85.71t}}{75} = -0.44e^{-85.71t}\ A$$



$V_L = V_{100\Omega} =$ توازی

$$-33e^{-85.71t}\ Volt$$

$$I_{100\Omega} = \frac{V}{R} = \frac{-33e^{-85.71t}}{100} = -0.33e^{-85.71t}\ A$$

$$V_{20\Omega} = I_{100\Omega}R = -0.33e^{-85.71t} \times 20$$

$$V_{20\Omega} = -6.6e^{-85.71t}\ Volt$$

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Al-Balqa' Applied University

جامعة البلقاء التطبيقية

Electrical circuits 1.

Final Exam
First semester
2016-2017,

Date: Jan. 7, 2016

Exam Time: 2hours

Choose the correct answer.

*** For the questions 1,2,3,4 consider the circuit shown in figure 1 and answer:**

1) The value of V_o is:

a) 20 V b) 10 V
c) 40 V d) -40 V

2) The value of I is:

a) 4A b) 1A
c) 2A d) -4A

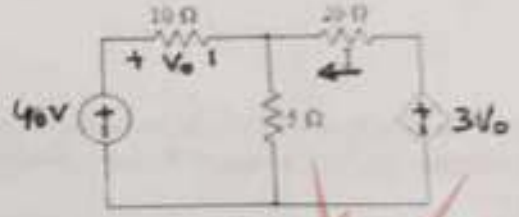


Fig.1

3) The power absorbed by 5 Ω resister is:

a) 100W b) 40W c) 160W d) 80W

4) The power delivered by 40V voltage source is:

a) 180W b) 40W c) 80W d) 100W

*** For the questions 5,6,7,8 consider the circuit shown in figure 2 and answer:**

5) The value of R_{TH} with respect to terminals (a,b) is:

a) 6Ω b) 3Ω
c) 14Ω d) 16Ω

6) The value of V_{TH} from the terminals (a,b) is:

a) 28V b) 12V
c) 32V d) 6V



Fig.2

7) The maximum power can be delivered to any load resister is:

a) 3W b) 9W c)36W d) 18W

8) The current I is:

a) 3A b) 1.5A c)4.5A d) 9A

*** For the questions 9,10,11,12 consider the circuit shown in figure 3 and answer:**

9) The value of *v(0)* is:

a) 20V b) 24V
c) 8V d) 16 V

10) the value of $v\ (t = \infty)$ is:

a) 8V b) 0V
c)24V d) 12 V

11) The value of *τ* (time constant) is:

a) 0.5s b)0.5 ms
c)2 ms d) 10 ms

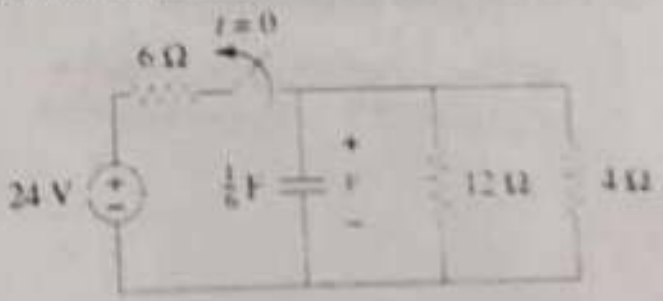


Fig.3

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12) The value of v at t = 750ms is:
a) 1.41V b) 0.8925 V c) 1.785V d) 3.57V

13) Refer to figure 4; the value of the equivalent capacitance is:
a) 80μF b) 40 μF c) 320 μF d) 170 μF

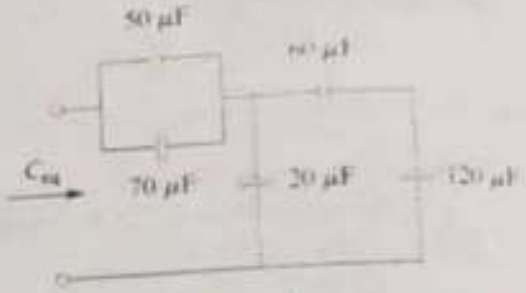


Fig 4

* For the questions 14,15,16 consider the circuit(the circuit under dc steady state conditions) shown in figure 5 and answer:

14)The inductor current i_L is:
a)2.5A b)10A
c)5A d)7.5A

15) The value of the capacitor voltage Vc is:
a) 10V b)20V
c) 15V d) 5V

16) The energy stored in the capacitor is:
a) 50 J b)450 J c) 200 J d) 800 J

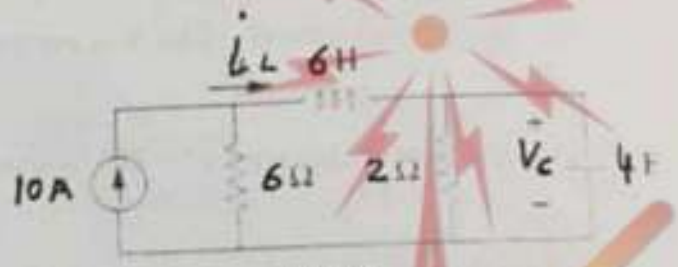


Fig.5

For the questions 17,18,19,20 consider the circuit shown in figure 6. The switch was open for a long time, but closed at t=0, If i(0)=10A. Answer:

17) The value of i(t=∞) is:
a) 6A b) 12A
c)1A d) 5A

18) The time constant is:
a)500ms b)400ms
c)200ms d) 20 ms

19) The value of i(t=100ms) is:
a)3A b)6A
c)4.2A d)8.4A

20) The complete response i(t) is:

a) $i(t)=12+6e^{-50t}$ A b) $i(t)=6+4e^{-5t}$ A
c) $i(t)=6+e^{-5t}$ A d) $i(t)=6+4e^{-0.5t}$ A

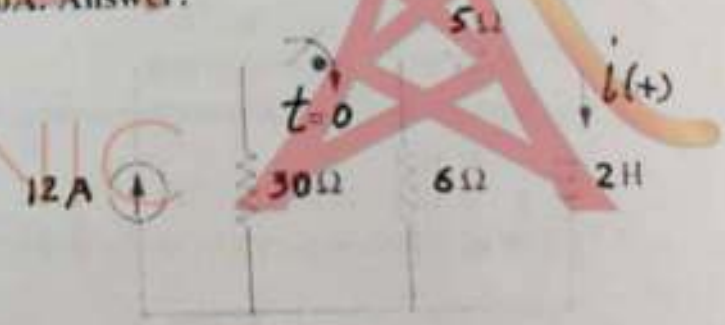


Fig.6

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* For the questions 21,22,23,24,25 consider the circuit shown in figure 7. Assuming v(0)=5V,i(0)=0,L=1H,C=10mF,R=1.923 Ω. Answer:



Fig.7

21) The value of α (damping factor)is:

a) 52 b) 6.5
c) 13 d) 26

22) The value of ω_o(resonant frequency) is:

a) 40 b)10
c)20 d)5

23) The value of S1 is:

a) -6 b)-4 c)-2 d) -25

24) The value of S2 is:

a) -100 b)-25 c) -12.5 d) -50

25) The natural response v(t) for t> 0 is:

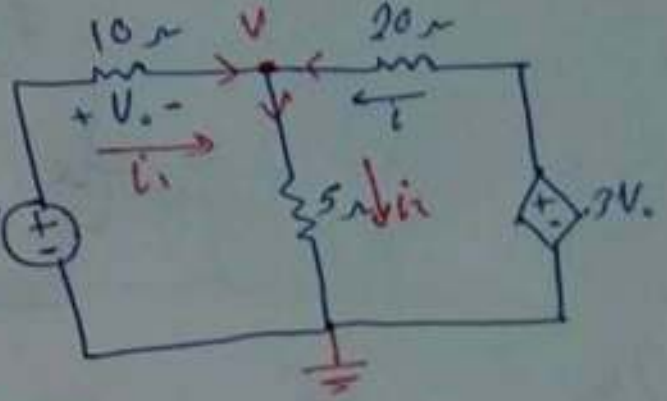
a) $v(t)= -0.2083e^{-2t} + 5.208e^{-50t}$ b) $v(t)= -0.2083e^{-4t} + 5.208e^{-25t}$
c) $v(t)= 0.2083e^{-2t} - 5.208e^{-50t}$ d) $v(t)= 0.2083e^{-4t} - 5.208e^{-25t}$

1	2	3	4	5	6	7	8	9	10	11	12	13
14	15	16	17	18	19	20	21	22	23	24	25	

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Q1



$$\frac{40-V}{10} + \frac{3(40-V)-V}{20} = \frac{V}{5}$$

$$80 - 2V + 120 - 3V - V - 4V = 0$$

$$10V = 200 \rightarrow \boxed{V = 20 \text{ Volt}}$$

$V_0 = 40 - V$

$$V_0 = 40 - 20 = \boxed{20 \text{ Volt}}$$

Q2 $$I = \frac{3V_0 - V}{20} = \frac{3 \times 20 - 20}{20} = \boxed{2 \text{ A}}$$

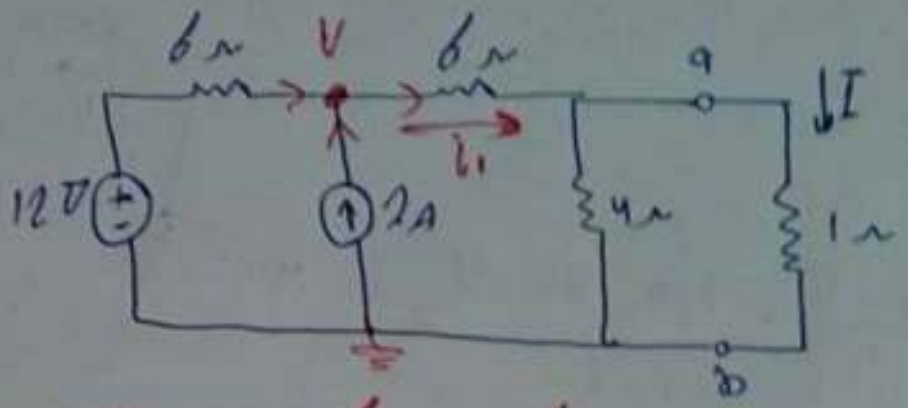
Q3 $$i_1 = \frac{40-20}{10} = 2 \text{ A}$$

$$i_2 = i_1 + i = 2 + 2 = \boxed{4 \text{ A}}$$

$$P = I^2 R = (4)^2 5 = \boxed{80 \text{ watt}}$$

Q4 $$P = VI = -40 \times 2 = -80 \text{ watt}$$

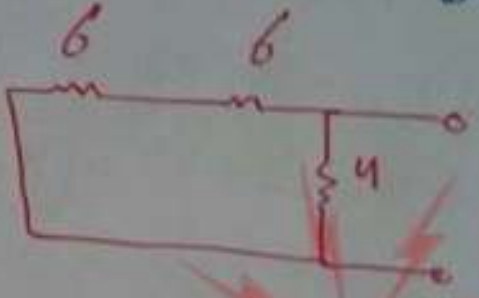
$$P_{delivered} = -(-80) = \boxed{80 \text{ watt}}$$



Q5

$R_{th} = [6\Omega \text{ series } 6\Omega] \,//\, 4\Omega$

$R_{th} = 3\Omega$



Q6

$$\frac{12 - V}{6} + 2 = \frac{V}{10} \rightarrow 8V = 12$$

$V = 1.5$ Volt

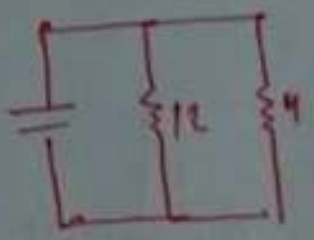
$$i_1 = \frac{V}{R} = \frac{1.5}{10} = 1.5A$$

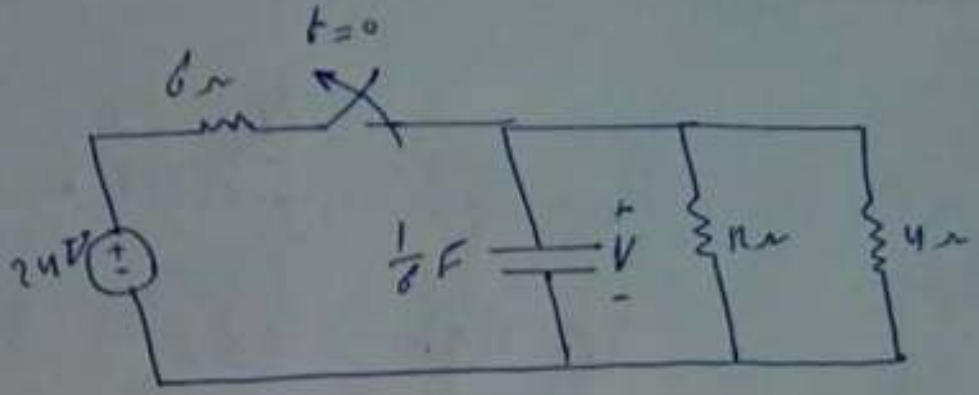
$$V_{th} = V_{4\Omega} = 1.5 * 4 = 6 \text{ Volt}$$

Q7 $$P_{max.} = \frac{V_{th}^2}{4 R_{th}} = \frac{6^2}{4(3)} = 3 \text{ Watt}$$

Q8 $$I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{6}{3+1} = \frac{6}{4} = 1.5 \text{ A}$$

$R_{th} = 3\,\Omega$

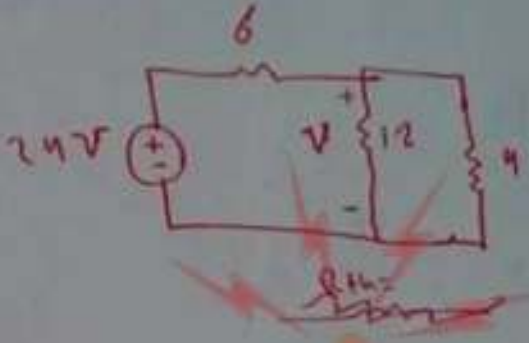




Q9

$$V_o = \frac{V * R_{eq}}{R_{eq} + R}$$

$$= \frac{24 * 3}{6 + 3} = \boxed{8 \text{ Volt}}$$



$R_{th} = 3\,\Omega$

$\tau = R_{th} C$

$= 3 * \frac{1}{6}$

$= 0.5\ s$

Q10 $V = V_o e^{-t/\tau} = 8e^{-\infty/0.5}$

$= 8e^{-\infty} = \frac{8}{\infty} = \boxed{0 \text{ Volt}}$

Q11 $\boxed{\tau = 0.5\ s}$

Q12 $V = V_o e^{-t/\tau} = 8e^{\frac{-75*10^{-3}}{0.5}} = \boxed{1.785 \text{ Volt}}$

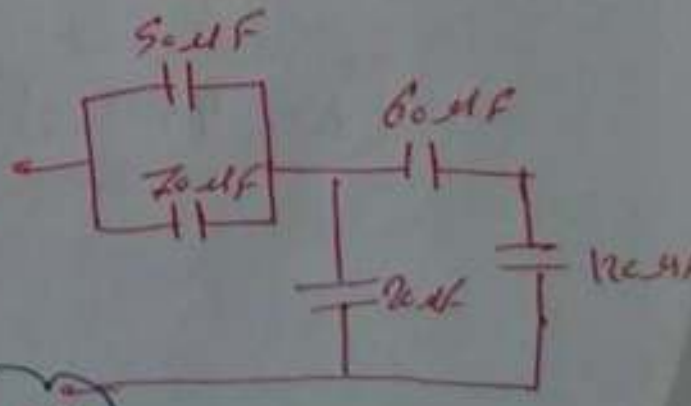
Q13

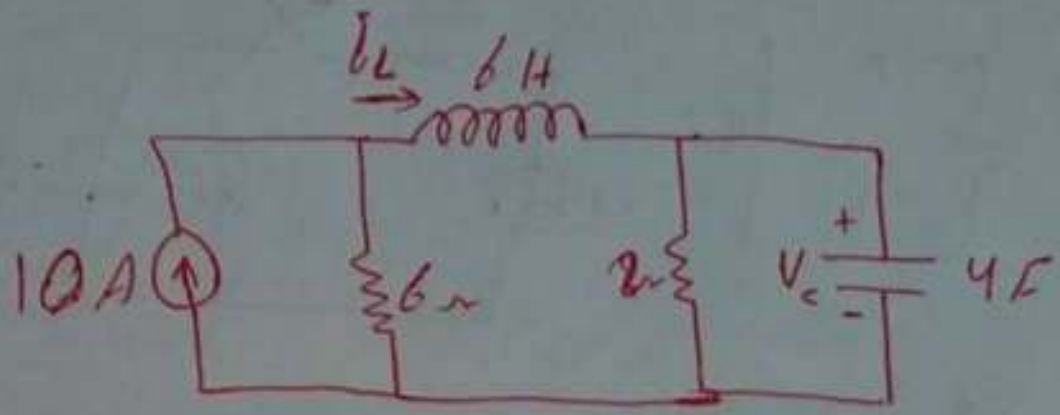
$[60\,\mu F \text{ series } 120\,\mu F] = 40\,\mu F$

$[40\,\mu F \,//\, 20\,\mu F] = 60\,\mu F$

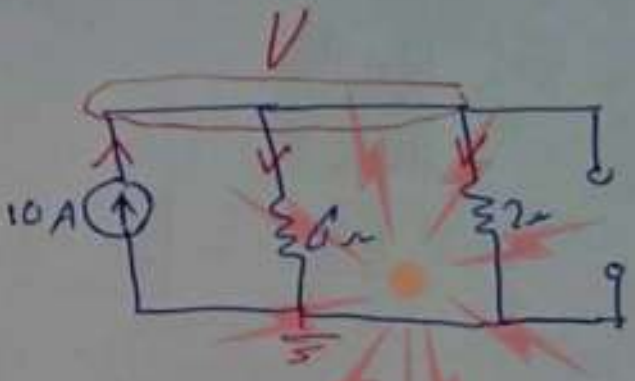
$[50\,\mu F \,//\, 70\,\mu F] = 120\,\mu F$

$[120\,\mu F \text{ series } 60\,\mu F] = \boxed{40\,\mu F}$





Q14



$$10 = \frac{V}{6} + \frac{V}{2}$$

$$\rightarrow V = 15 \text{ Volt}$$

$$i_L = i_{2\Omega} = \frac{V}{2} = \frac{15}{2} = 7.5 \text{ A}$$

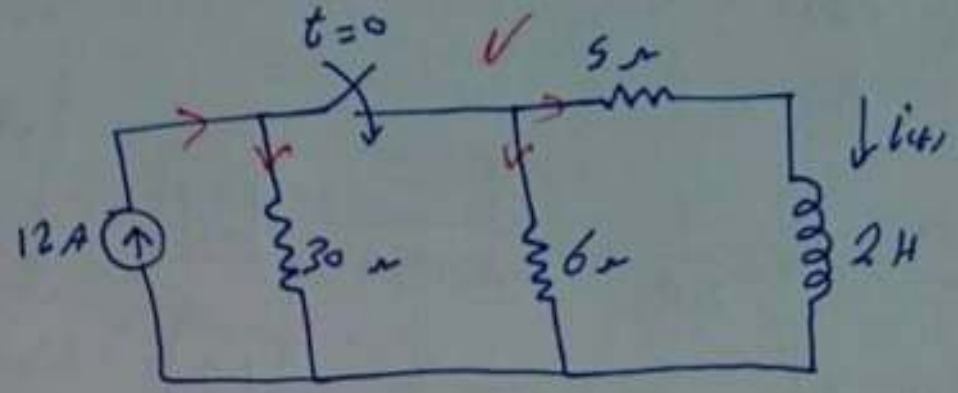
(توازی)

Q15 $V_c = V = 15$ Volt (توازی)

Q16 $$W = \frac{1}{2} C V^2 = \frac{1}{2} \times 4 \times (15)^2$$

$$W = 450 \text{ J}$$

Q17

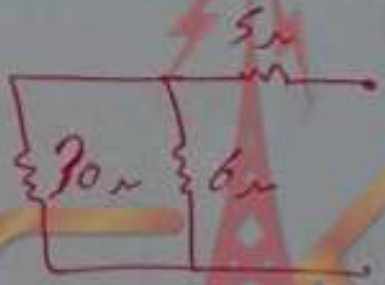


$$12 = \frac{V}{30} + \frac{V}{6} + \frac{V}{5} \longrightarrow \boxed{V = 30\ \text{Volt}}$$

$$i(t) = \frac{V}{R} = \frac{30}{5} = \boxed{6\ A}$$

$$\boxed{i(\infty) = i(6) = 6\ A}$$

Q18



$R_{th} = (30 \,//\, 6)$ series 5

$$\boxed{R_{th} = 10\ \Omega}$$

$$T = \frac{L}{R_{th}} = \frac{2}{10} = 0.2 = \boxed{200\ ms}$$

Q19

$$i(100\ ms) = i(\infty) + (i_0 - i(\infty))\, e^{-\frac{100\ m}{0.2}}$$

$$= 6 + (10 - 6)\, e^{-\frac{100\ m}{0.2}} = \boxed{8.4\ A}$$

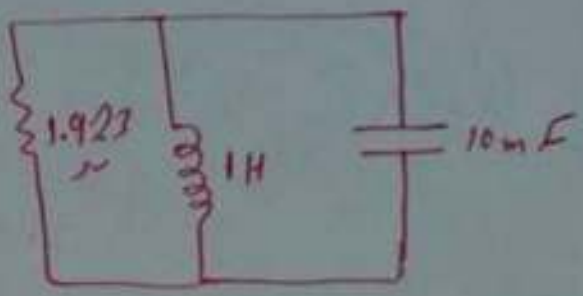
Q20

$$i(t) = i(\infty) + (i_0 - i_\infty)\, e^{-6/t}$$

$$\boxed{i(t) = 6 + 4e^{-5t}}$$

$V(0) = 5$ Volt

$i(0) = 0$



Q21 $\alpha = \frac{1}{2RC} = \frac{1}{2 * 1.923 * 10m} = \boxed{26}$

Q22 $\omega_o = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 * 10m}} = \boxed{10}$

Q23 $S_1 = -\alpha + \sqrt{\alpha^2 - \omega_o^2} = -26 + \sqrt{26^2 - 10^2} = \boxed{-2}$

Q24 $S_2 = -\alpha - \sqrt{\alpha^2 - \omega_o^2} = -26 - \sqrt{26^2 - 10^2} = \boxed{-50}$

Q25 $\alpha > \omega \rightarrow$ over damped

$V(t) = A_1 e^{S_1 t} + A_2 e^{S_2 t}$

$V(t) = A_1 e^{-2t} + A_2 e^{-50t}$

$V(0) = A_1 + A_2 \rightarrow \boxed{5 = A_1 + A_2}$ — ①

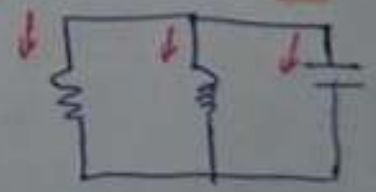
POLYTECHNIC

$\frac{dV}{dt} = -2A_1 e^{-2t} - 50A_2 e^{-50t}$

عوض (t=0) في المعادلة

$\frac{dV(0)}{dt} = -2A_1 - 50A_2$

$\boxed{0 = -2A_1 - 50A_2}$ — ②



$0 = i_R + i_L + i_C$

$0 = \frac{V}{1.923} + 0 + 10m \frac{dV}{dt}$

عوض (0) في المعادلة

$\boxed{\frac{dV(0)}{dt} = 0}$

$\boxed{A_1 = 5.2}$ $\boxed{A_2 = -0.2}$

$\boxed{V(t) = 5.2e^{-2t} - 0.2e^{-50t}}$ → الجواب النهائي

Al-Balqa' Applied University
Faculty of
EngineeringTechnology
Electrical circuits 1



جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجيه

Final exam 4-08-2015

Q1: Find the voltage (v) for the network shown in Fig 1 using superposition theorem. (10marks)

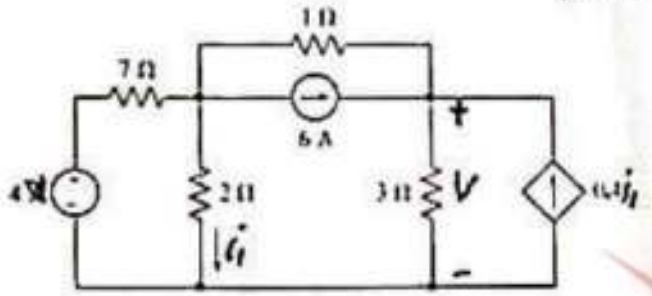


Fig 1

Q2: For the circuit shown in Fig 2: (10marks)
a) Determine the value of Rl to which a maximum power can be delivered .
b) The maximum power deliverd to Rl.

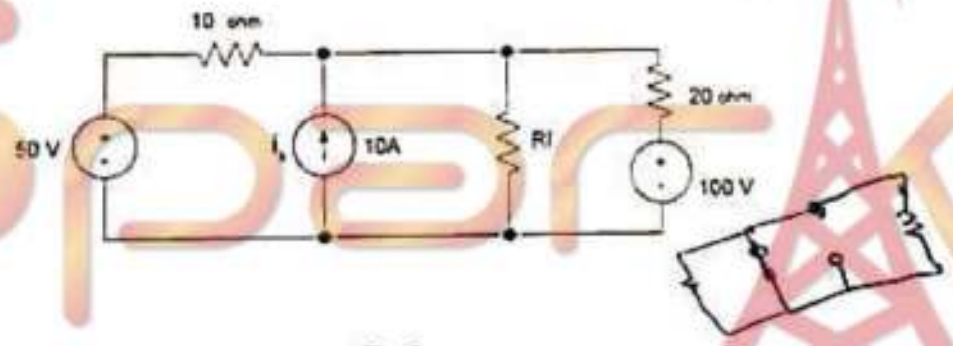


Fig 2

Q3: a) For the circuit shown in Fig 3.a find the equivalent capacitance (10 marks)

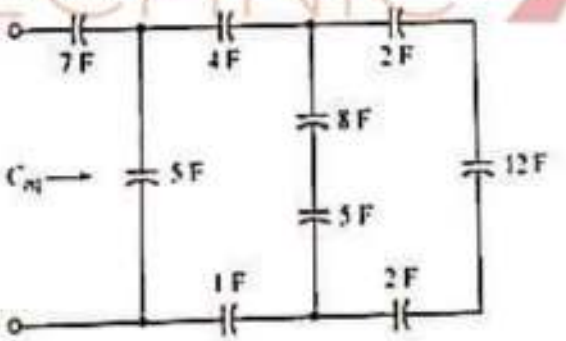


Fig 3.a

b) Calculate the power dissipated in the 40 Ω resistor and the voltage labeled Vc for the circuit given in fig 3.b

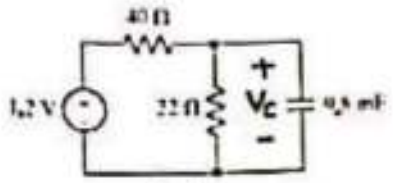


fig 3.b

Q4: The voltage across a 33 mF capacitor is shown graphically in Fig 4 : (8 marks)

a) Assuming the passive sign convention, sketch the resulting current waveform across the capacitor.

b) Find the energy stored in the capacitor at t=20ms.



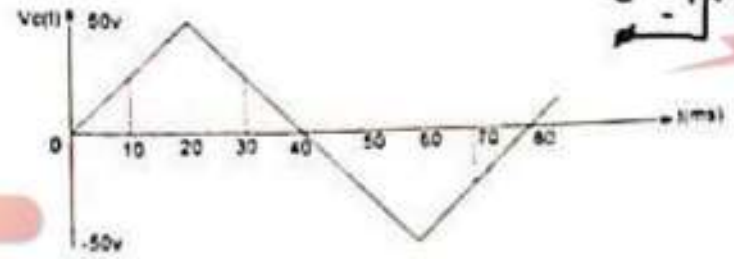


Fig 4

Q5: For the circuit shown in Fig 5 the switch opens at t=0 after having been closed for a long time. Find. (12 marks)

a) $i_1(t)$ for $t >= 0^-$.
b) $i_L(t)$ for $t >= 0^-$.
c) $V_2(t)$ for $t >= 0^-$.

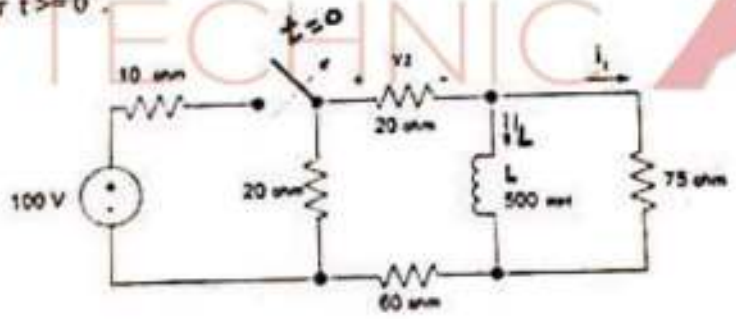


Fig 5

Good Luck

[امتحان فاينل
Circuit 1]

Sultan

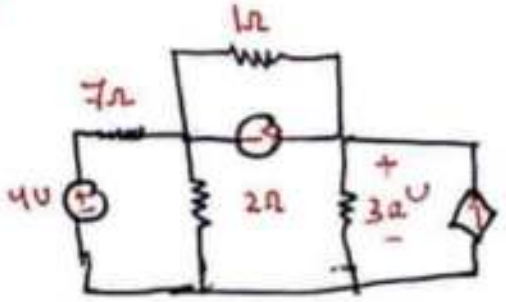
4/8/2015

Q1

$v = v' + v''$

$= 0.69 + 2.682$

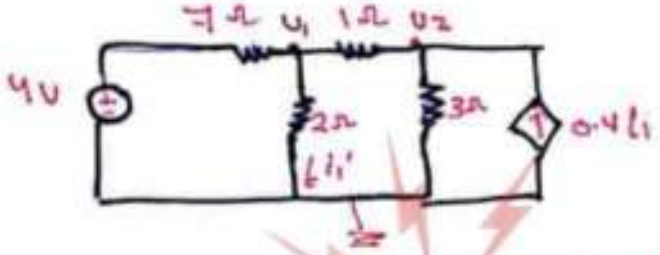
$v = 3.372\ V$



node (v_1)

$$\frac{4 - v_1}{7} = \frac{v_1}{2} + v_1 - v_2$$

$$23v_1 - 14v_2 = 8$$

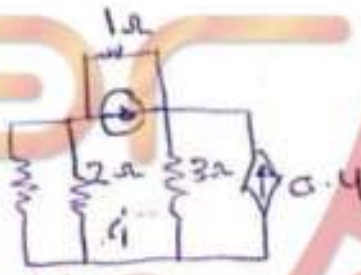


node (v_2)

$$v_1 - v_2 + 0.4\frac{v_1}{2} = \frac{v_2}{3} \rightarrow 7.2v_1 - 8v_2 = 0$$

$$v_1 = 0.76\ V \qquad v' = v_2 = 0.69\ \text{volt}$$

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#spark-team



→ Mesh ①

$$I_1 = 0.4i_1' = 0.4(I_2 - I_3)$$

→ Mesh (2)

$$6I_2 - 2I_3 - 3I_1 = -6$$

$$6I_2 - 2I_3 - 3(0.4(I_2 - I_3)) = -6$$

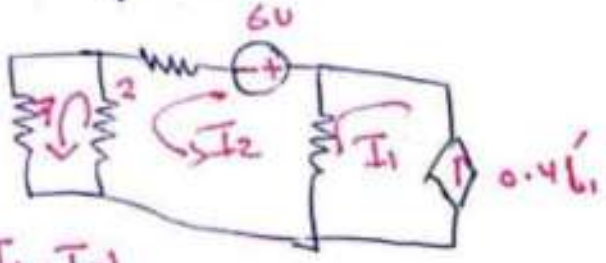
$$\boxed{4.8I_2 - 0.8I_3 = -6}$$

→ Mesh ③

$$-2I_2 + 9I_1 = 0$$

$I_2 = -1.298\ A$

$I_3 = -0.288\ A$



$$v'' = 3(I_1 - I_2)$$

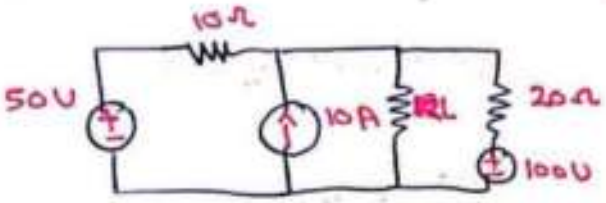
$$v'' = 3(-0.404 + 1.298)$$

$$v'' = 2.682\ V$$

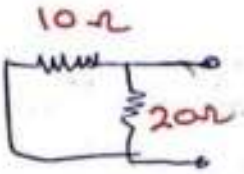
$$I_1 = 0.4(-1.298 + 0.288)$$

$$\boxed{I_1 = -0.404}$$

Q.2

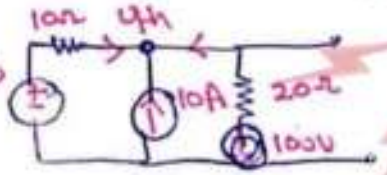


$R_{Th} = R_L = 6.667\ \Omega$



→ Node (U_{Th})

$$\frac{50 - U_{Th}}{10} + 10 + \frac{100 - U_{Th}}{20} = 0$$



$100 - 2U_{Th} + 200 + 100 \rightarrow U_{Th} = 0$

$3U_{Th} = 400 \rightarrow U_{Th} = 133.34$ Volt

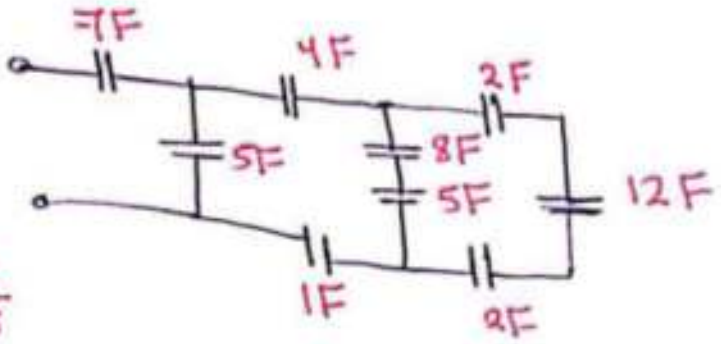
$$P_{max} = \frac{U_{Th}^2}{4R_{Th}} = \frac{(133.34)^2}{4 \times 6.667}$$

$P_{max} = 666.7$ Watt

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POLYTECHNIC

Q3 a)



[2 series 12 series 2] → $\frac{12}{13}$

[8 series 5] → $\frac{40}{13}$

[$\frac{12}{13}$ // $\frac{40}{13}$] → 4

[4 series 4 series []] → $\frac{2}{3}$

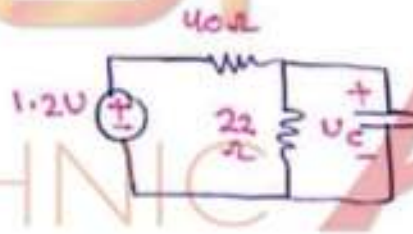
[$\frac{2}{3}$ // 5] → $\frac{17}{3}$

C_{eq} = [$\frac{17}{3}$ series 7]

$C_{eq} = 3.13\,F$

#spark-team

Q3 b)



$U_{40\Omega} = \frac{1.2 \times 40}{40+22}$

$= \frac{24}{31}$ volt.

$R_{40} = \frac{U^2}{R} = 0.0149$ Watt.

$U_c = U_{22\Omega} = \frac{1.2 \times 22}{22+40} = 0.425$ volt.

POLYTECHNIC

Q4| a

step①

$$= \frac{50-0}{20-0} = 2.5$$

$$y - y_0 = S_e(x - x_0) \dashrightarrow y - 50 = 2.5(x-20) \dashrightarrow y = 2.5x$$

step②

$$= \frac{-50-50}{60-20} = -2.5$$

$$y + 50 = -2.5(x-60) \rightarrow \boxed{y = 100 - 2.5x}$$

step③

$$= \frac{0+50}{80-60} = 2.5$$

$$y - 0 = 2.5(x-80) \dashrightarrow \boxed{y = 2.5x - 200}$$

$$v(t) = \begin{cases} 2.5t & , 0<t<20 \\ 100-2.5t & , 20<t<60 \\ 2.5t-200 & , 60<t<80 \end{cases}$$

$$i(t) = C\frac{dv}{dt}$$

$$i_1(t) = 33m(2.5) = 82.5mA$$

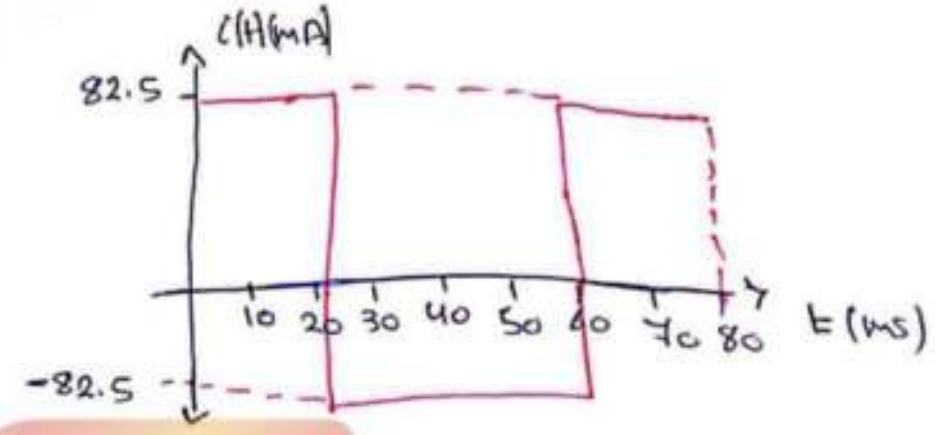
$$i_2(t) = 33m(-2.5) = -82.5mA$$

$$i_3(t) = 33m(2.5) = 82.5mA$$

SPARK POLYTECHNIC

Q4)

$$i(t) = \begin{cases} 82.5\ mA & , 0 < t < 20 \\ -82.5\ mA & , 20 < t < 60 \\ 82.5\ mA & , 60 < t < 80 \end{cases}$$



(b)

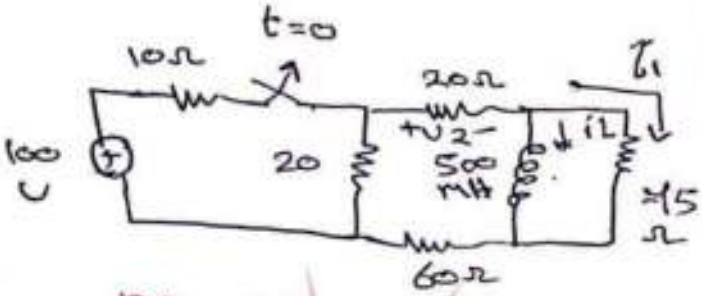
$$W = \frac{1}{2} C v^2$$

$$W = \frac{1}{2}(33m)(50)^2 = 41.25\ J$$

Sultanabdulaab

Spark # spark-team POLYTECHNIC

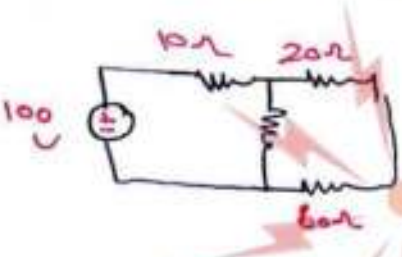
Q5]

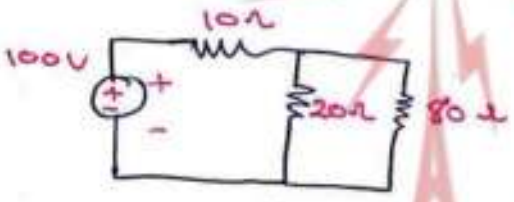


$$V_{16\Omega} = \frac{100 * 16}{16+10} = 61.53\ V$$

$$I_{L_0} = I_{8\Omega} = \frac{V}{R} = \frac{61.53}{80}$$

$$\boxed{I_{L_0} = 0.769\ A}$$

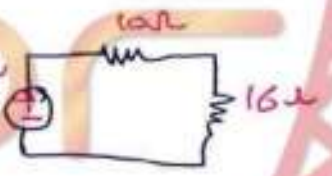


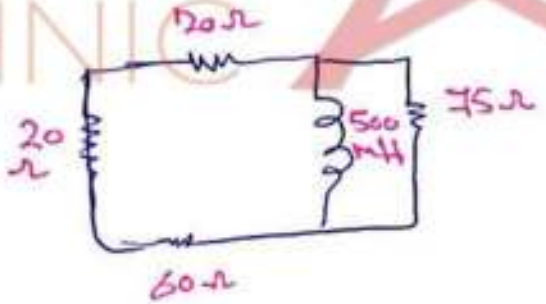


$$\boxed{R_{Th} = 42.85\ \Omega}$$

$$\tau = \frac{L}{R_{Th}} = \frac{500m}{42.85}$$

$$\boxed{= 0.01185}$$





$$I_L = I_{L(0)} e^{+t/\tau}$$

$$I_L = 0.769\, e^{+t/0.0118}$$

$$I_L = 0.769\, e^{-85.71t}$$

POLYTECHNIC

NAME : FHAMEES
FAX : 00962-6-334
TEL :
SEP.# : E71092F01763720

DATE,TIME 29/10 14:24
FAX NO./NAME 414
DURATION 00:00:21
PAGE(S) 02
RESULT OK
MODE STANDARD ECM

Q5

$$U_L = L\frac{dI}{dt} = 0.5\left(0.769t - 85.71e^{-85.71t}\right)$$

$$U_L = U_{75\Omega} = -33e^{-85.71t}$$

$$I = -0.44e^{-85.71t} \text{ A}$$



$U_L = U_{100\Omega}$

$= -33e^{-85.71t}$ volt

$$I_{100\Omega} = \frac{U}{R} = \frac{-33e^{-85.71t}}{100}$$

$$= -0.33e^{-85.71t} \text{ A}$$

$$U_{20\Omega} = I_{100\Omega}R = -0.33e^{-85.71t} * 20$$

$$U_{20\Omega} = -6.6e^{-85.71t} \text{ volt}$$

SPOT POLYTECHNIC

Al-Balqa' Applied University

جامعة البلقاء التطبيقية

Electrical circuits 1.

Final Exam
First semester
2016-2017.

Date: Jan. 7, 2016

Exam Time: 2hours

Choose the correct answer.

* For the questions 1,2,3,4 consider the circuit shown in figure 1 and answer:

1) The value of V_o is:
   a) 20 V b) 10 V
   c) 40 V d) -40 V

2) The value of I is:
   a) 4A b) 1A
   c) 2A d) -4A

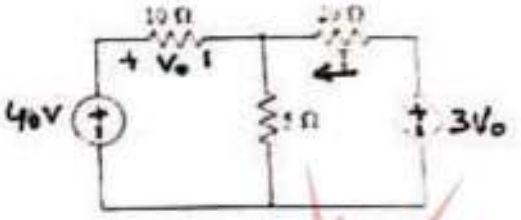


Fig.1

3) The power absorbed by 5 Ω resister is:
   a) 100W b) 40W c) 160W d) 80W

4) The power delivered by 40V voltage source is:
   a) 180W b) 40W c) 80W d) 100W

* For the questions 5,6,7,8 consider the circuit shown in figure 2 and answer:

5) The value of R_{TH} with respect to terminals (a,b) is:
   a) 6Ω b) 3Ω
   c) 14Ω d) 16Ω

6) The value of V_{TH} from the terminals (a,b) is:
   a) 28V b) 12V
   c) 32V d) 6V



Fig.2

7) The maximum power can be delivered to any load resister is:
   a) 3W b) 9W c)36W d) 18W

8) The current I is:
   a) 3A b) 1.5A c)4.5A d) 9A

* For the questions 9,10,11,12 consider the circuit shown in figure 3 and answer:

9) The value of $v(0)$ is:
   a) 20V b) 24V
   c) 8V d) 16 V

10) the value of $v(t=\infty)$ is:
   a) 8V b) 0V
   c)24V d) 12 V

11) The value of r (time constant) is:
   a) 0.5s b)0.5 ms
   c)2 ms d) 10 ms

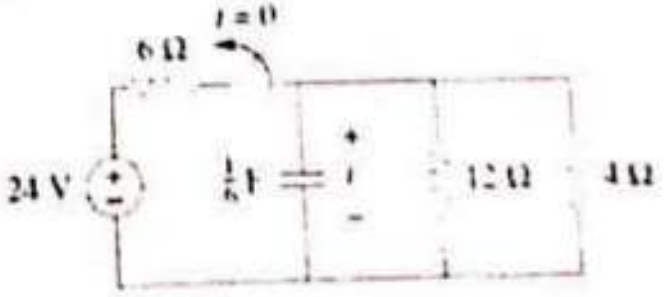


Fig.3

12) The value of v at t = 750ms is:
a) 1.41V b) 0.8925 V c) 1.785V d) 3.57V

13) Refer to figure 4; the value of the equivalent capacitance is:
a) 80μF b) 40 μF c) 320 μF d) 170 μF

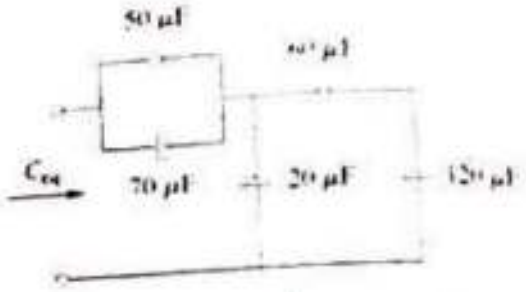


Fig 4

* For the questions 14,15,16 consider the circuit(the circuit under dc steady state conditions) shown in figure 5 and answer:

14)The inductor current i_L is:
a)2.5A b)10A
c)5A d)7.5A

15) The value of the capacitor voltage Vc is:
a) 10V b)20V
c) 15V d) 5V

16) The energy stored in the capacitor is:
a) 50 J b)450 J c) 200 J d) 800 J

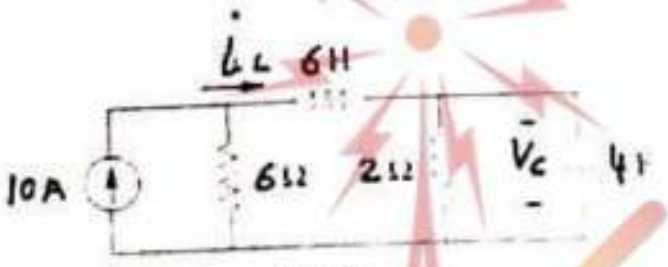


Fig.5

For the questions 17,18,19,20 consider the circuit shown in figure 6. The switch was open for a long time, but closed at t=0, If i(0)=10A. Answer:

17) The value of i(t=∞) is:
a) 6A b) 12A
c)1A d) 5A

18) The time constant is:
a)500ms b)400ms
c)200ms d) 20 ms

19) The value of i(t=100ms) is:
a)3A b)6A
c)4.2A d)8.4A

20) The complete response i(t) is:

a) $i(t)=12+6e^{-50t}$ A b) $i(t)=6+4e^{-5t}$ A
c) $i(t)=6+e^{-5t}$ A d) $i(t)=6+4e^{-0.5t}$ A

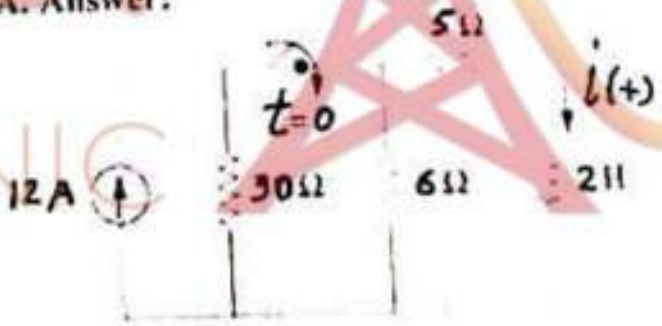


Fig.6

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* For the questions 21,22,23,24,25 consider the circuit shown in figure 7. Assuming v(0)=5V,i(0)=0,L=1H,C=10mF,R=1.923 Ω. Answer:

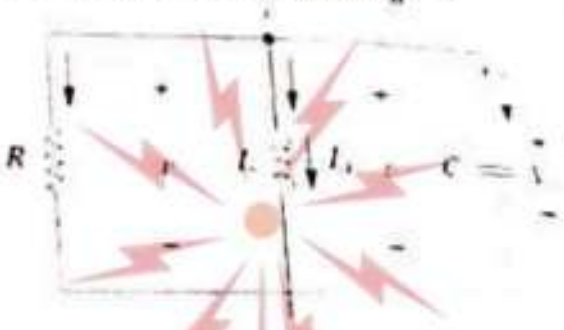


Fig.7

21) The value of α (damping factor)is:

a) 52 b) 6.5
c) 13 d) 26

22) The value of ω_o(resonant frequency) is:

a) 40 b)10
c)20 d)5

23) The value of S1 is:

a) -6 b)-4 c)-2 d) -25

24) The value of S2 is:

a) -100 b)-25 c) -12.5 d) -50

25) The natural response v(t) for t> 0 is:

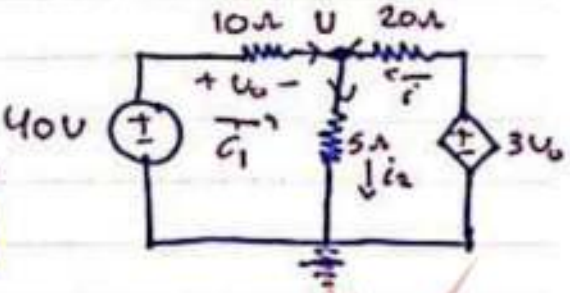
a) $v(t)= -0.2083e^{-2t} + 5.208e^{-50t}$ b) $v(t)= -0.2083e^{-4t} + 5.208e^{-25t}$
c) $v(t)= 0.2083e^{-2t} - 5.208e^{-50t}$ d) $v(t)= 0.2083e^{-4t} - 5.208e^{-25t}$

1	2	3	4	5	6	7	8	9	10	11	12	13
14	15	16	17	18	19	20	21	22	23	24	25	

Good Luck

No

$$\frac{40-U}{10} + 3\left(\frac{40-U}{20}\right) - U = \frac{U}{5}$$



$$80 - 2U + 120 - 3U - U - 4U = 0$$

$$10U = 200 \rightarrow \boxed{U = 20 \text{ volt}}$$

$$\therefore V_o = 40 - 20$$

$$\boxed{= 20 \text{ volt}}$$ A

Q2 $$I = \frac{3V_o - U}{20} = \frac{3 \times 20 - 20}{20}$$

$$\boxed{= 2A}$$ C

Q3 $$i_1 = \frac{40-20}{10}$$

$$= 2A$$

$$i_2 = i_1 + i = 2 + 2 = 4A$$

$$P = I^2R = (4)^2 5 = \boxed{80 \text{ watt}}$$ b

Q4 $$P = VI$$

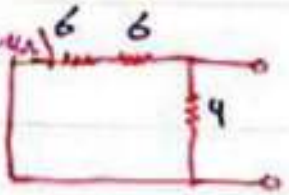
$$= -40 \times 2 = -80 \text{ watt}$$

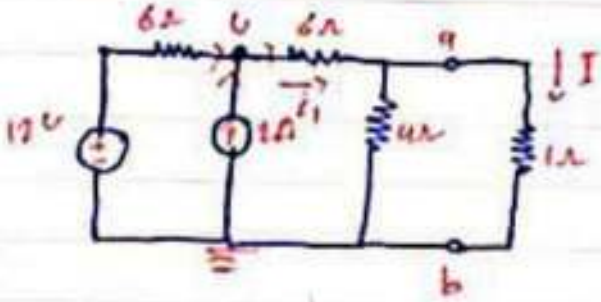
$$P_{delivered} = -(-80) = \boxed{80 \text{ watt}}$$ C

Q5

R_{th} = [6Ω series 6Ω] // 4Ω

$R_{th} = 3\Omega$





b

Q6

$$\frac{12-U}{6} + 12 = \frac{U}{10}$$

$$\rightarrow 8U = 120 \therefore U = 15 \text{ volt}$$

$$i_1 = \frac{U}{R} = \frac{15}{10} = 1.5A$$

$$\rightarrow U_{th} = U_{4\Omega} = 1.5 \times 4 = 6 \text{ volt}$$ d

Q7 $P_{max} = \frac{U_{th}^2}{4R_{th}} = \frac{6^2}{4(3)} = 3 \text{ watt}$ c

Q8

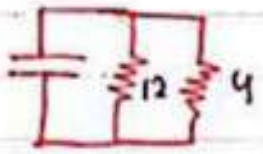
$$I_L = \frac{U_{th}}{R_{th}+R_L} = \frac{6}{3+1} = \frac{6}{4} \Rightarrow 1.5A$$ b

Sultan

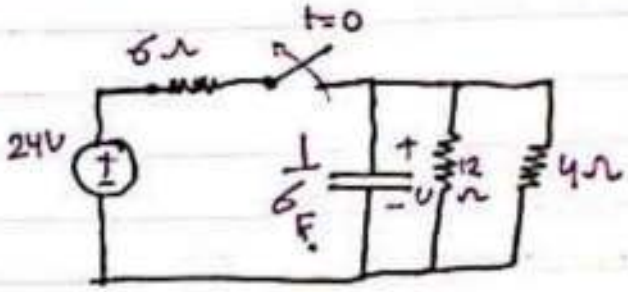
فائنل سیریز 1

No

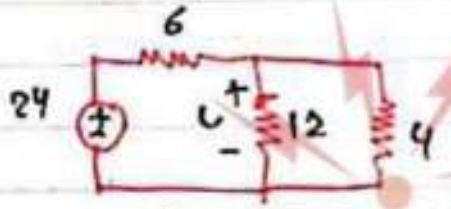




* $R_{th} = 3\Omega$



Q9 $V_o = \frac{V * R_{eq}}{R_{eq} + R}$



$= \frac{24 * 3}{6+3} = 8$ Volt

Q10 $V = V_o e^{-t/\tau} = 8e^{-\infty/0.5}$

$= 8e^{-\infty} = \frac{8}{\infty} = 0$ Volt

$\tau = R_{th} C$

$= 3 * \frac{1}{6}$

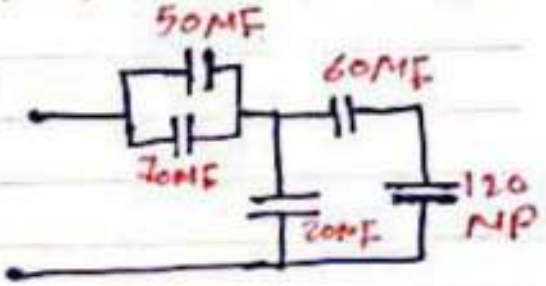
$= 0.5$ s

Q11 $\tau = 0.5$ s

Q12 $V = V_o e^{-t/\tau} = 8e^{-\frac{750 \times 10^{-3}}{0.5}}$

$= 1.785$ Volt

Q13)



[60MF Series 120MF] = 40MF

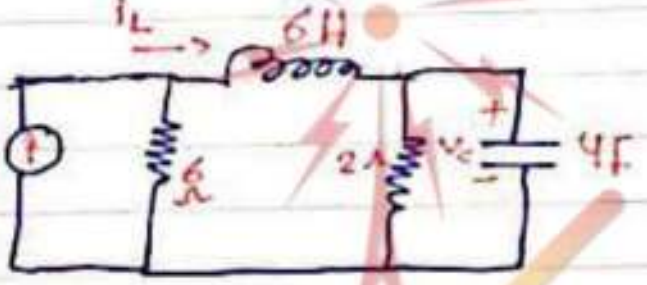
[40MF // 20MF] = 60MF

[50MF // 70MF] = 120MF

[120MF Series 60MF] = 40MF

∴ b

Q14

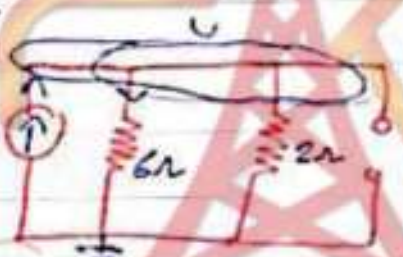


$$10 = \frac{v}{6} + \frac{v}{2}$$

$$\therefore v = 15 \text{ Volt}$$

$$i_L = i_{2\Omega} = \frac{v}{a} = \frac{15}{2} = 7.5A$$

∴ a



Q15) $V_c = v = 15$ Volt

∴ c

Q16 $w = \frac{1}{2} C v^2 = \frac{1}{2} * 4 (15)^2$

$$\therefore w = 450 J$$

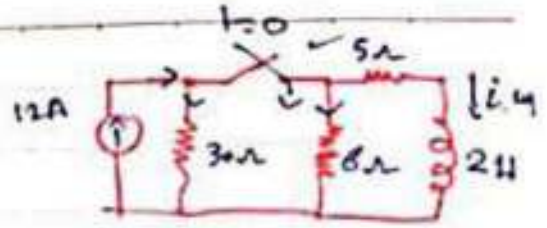
∴ b

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Q17)

$$12 = \frac{v}{30} + \frac{v}{6} + \frac{v}{5}$$

$\therefore$ $v = 30$ Volt



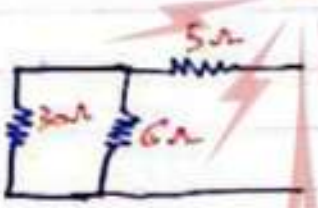
$$i(t) = \frac{v}{R} = \frac{30}{5} = 6A$$ (a)

$i(\infty) = i(0) = 6A$

Q18

$R_{th} = (30 \| 6)$ series

$R_{th} = 10\Omega$



$$\tau = \frac{L}{R_{th}} = \frac{2}{10} = 0.2 = 200ms$$ c

Q19) $i(100ms) = i(\infty) + (i_0 - i(\infty))e^{-t/\tau}$

$= 6 + (10 - 6)e^{\frac{-100m}{0.2}} = 8.4A$

d

Q20

$i(t) = i(\infty) + (i_0 - i_\infty)e^{-t/\tau}$

$i(t) = 6 + 4e^{-5t}$ b

No

$u(b) = 5u \cdot H$ $\quad R = 1.923\,\Omega$

$i(0) = 0$ $\quad L = 1H$

$u(0) = 5V$



Q21 $\alpha = \frac{1}{2RC} = \frac{1}{2 \times 1.923 \times 10m} = 26$ d

Q22) $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 10m}} = 10$ b

Q23) $S_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -26 + \sqrt{26^2 - 10^2} = -2$ c

Q24) $S_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -26 - \sqrt{26^2 - 10^2} = -50$ a

No

$u(b) = 5u \cdot H$ $\quad R = 1.923\,\Omega$

$i(0) = 0$ $\quad L = 1H$

$u(0) = 5V$



Q21 $\alpha = \frac{1}{2RC} = \frac{1}{2 \times 1.923 \times 10m} = 26$ d

Q22) $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 10m}} = 10$ b

Q23) $S_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -26 + \sqrt{26^2 - 10^2} = -2$ c

Q24) $S_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -26 - \sqrt{26^2 - 10^2} = -50$ a

No

Q25) $\alpha > \omega$ → over damped

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$v(t) = A_1 e^{-2t} + A_2 e^{-50t}$$

$$v(0) = A_1 + A_2 \longrightarrow 5 = A_1 + A_2 \quad \text{— (1)}$$

$$\frac{dv}{dt} = -2A_1 e^{-2t} - 50A_2 e^{-50t}$$

$$\frac{dv(0)}{dt} = \frac{-v_0 - Ri_0}{RC}$$

$$= \frac{-5 - R \times 0}{1.923 \times 10 \times 10^{-3}} = -260$$

$$-260 = -2A_1 - 50A_2 \quad \text{— (2)}$$

$$-260.01 = 2A_1 - 50A_2$$

$$5 = A_1 + A_2$$

$$\rightarrow \omega = 5.2e^{-2t} + 0.2e^{-50t}$$

$$\frac{dv(0)}{dt} = -2A_1 - 50A_2$$

$$-260 = -(R + L + C$$

Q1. Use superposition theorem to find i_0 and the power absorbed by the 6Ω resistor in the circuit shown in figure 1.

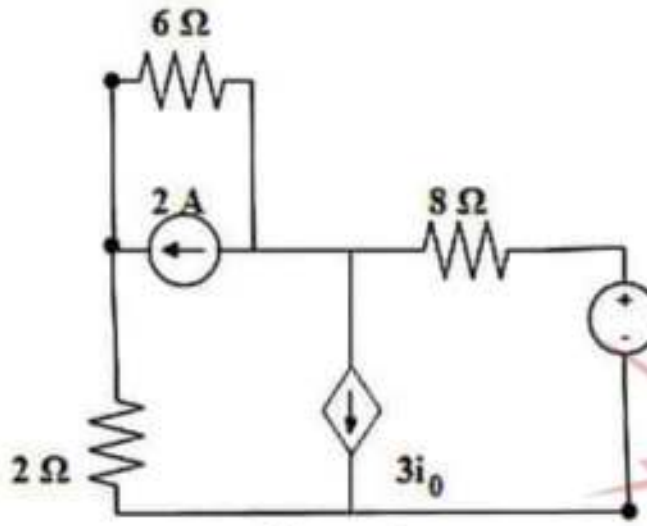


Figure 1.

Q.2 In the circuit of Figure 2:

a) **Find R so that maximum power is transferred to the resistance R.**
b) **Find this maximum power.**

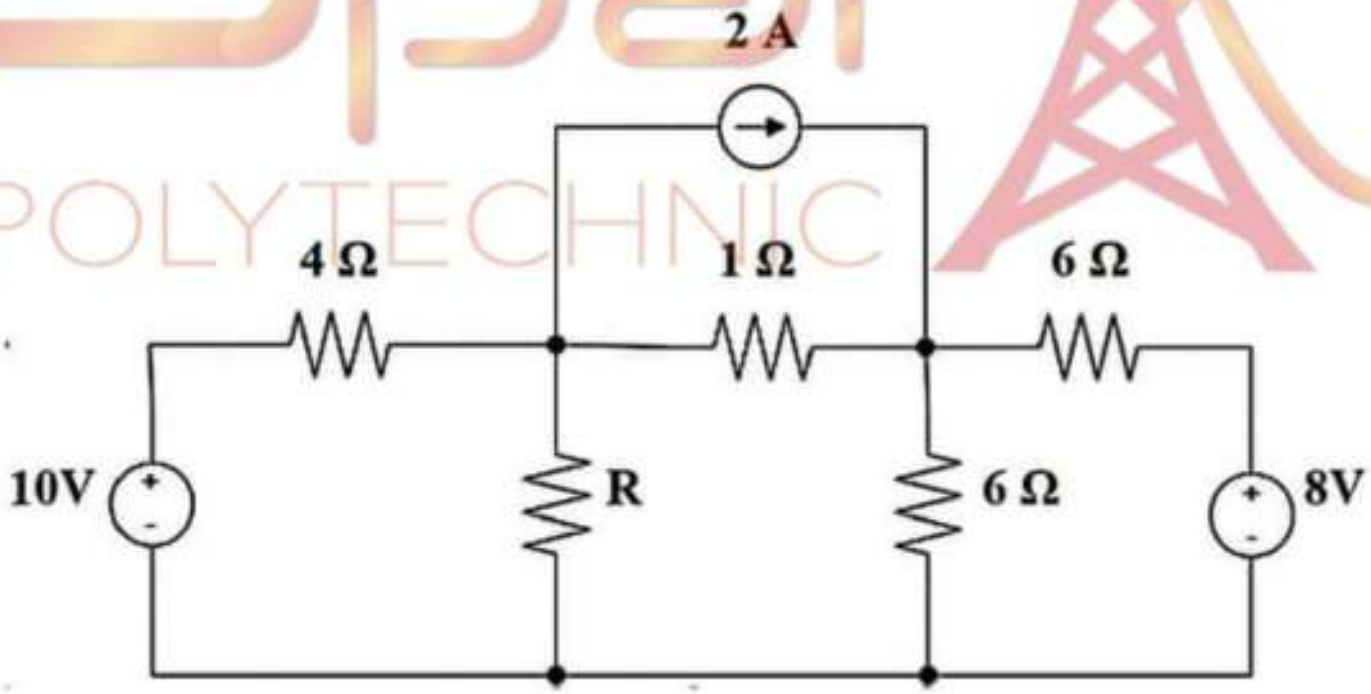


Figure 2.

For the question 1, 2 and 3 consider the circuit shown in figure 3 under dc steady state conditions and answer:

1) The capacitor voltage is:

a. 10V b.5V c. 20V d. 60V

2) The energy stored in the capacitor is:

a. 10j b.5j c. 90j d. 40j

3) The current i_x is:

a. 12A b. 4A c. 8A d. 3A

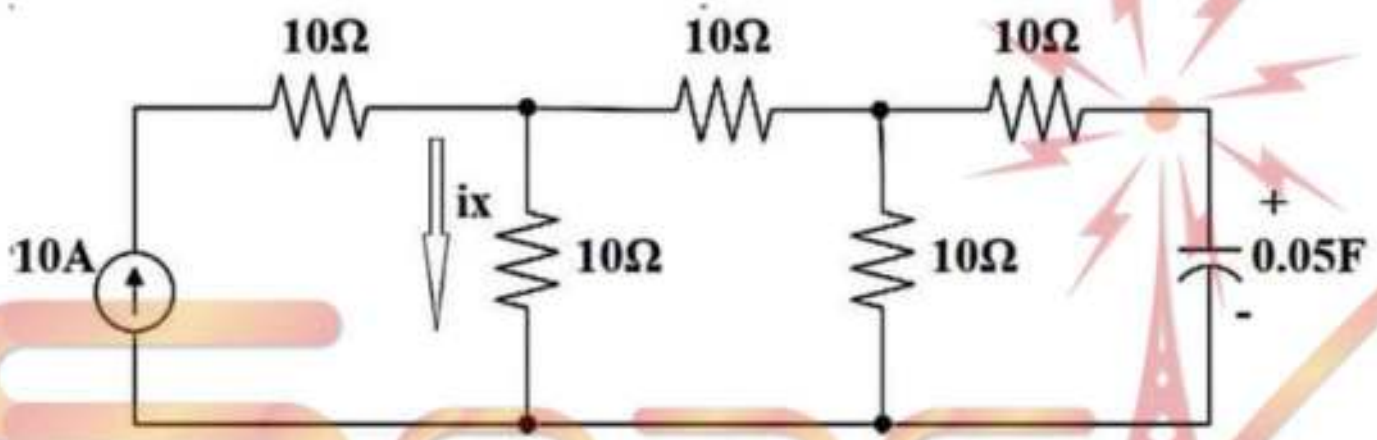


Figure 3.

For the questions 4, 5 and 6 consider the circuit shown in figure 4. Use Norton's theorem with respect to terminals (a-b) and answer:

4) The Norton current is:

a. 2A b. 6A c. 3A d. 4A

5) The Norton resistance is:

a. 0.75Ω b. 3Ω c. 1.5Ω d. 6Ω

6) The current I is:

a. 1.5A b. 4A c. 3A d. 6A

For the question 7 and 8 consider the current in a 5H inductor is i(t)=10(1-3t) A. Assume V (0) = 0 V. Answer:

7) The voltage at the terminals of the inductor at t = 3s is:

a. 150V b. -150V c. 350V d. -350V

8) The energy stored in the inductor at t = 1s is:

a. -320j b. 250j c. 1000j d. 500j

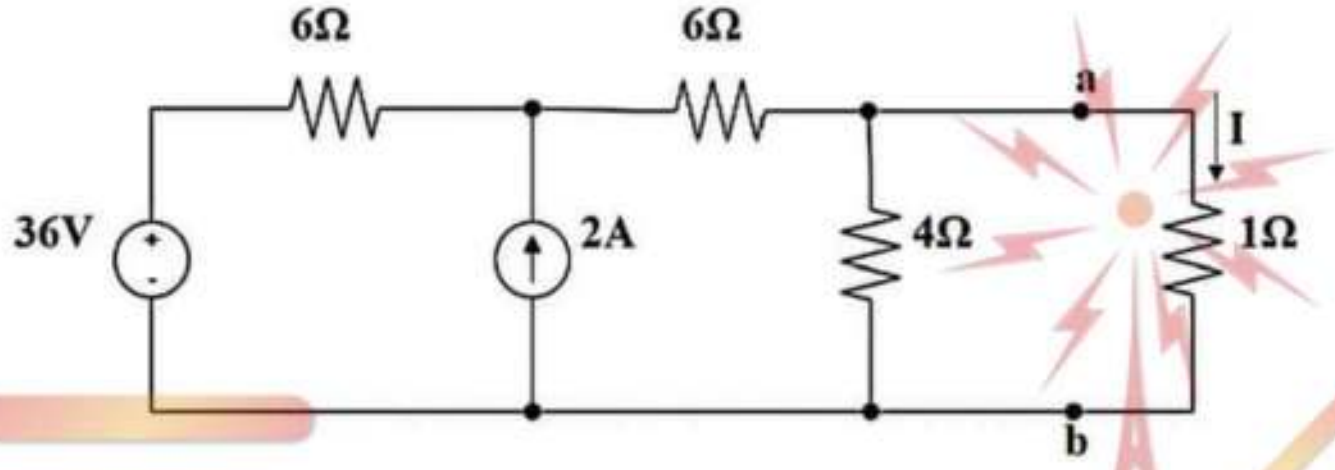


Figure 4.

The switch in Figure 5 has been closed for a long time. It opens at t = 0.

9) The steady state response i(t) is .....

10) The transient response i(t) is .....

11) The complete response is .....

12) The inductor current at t = 200 ms is .....

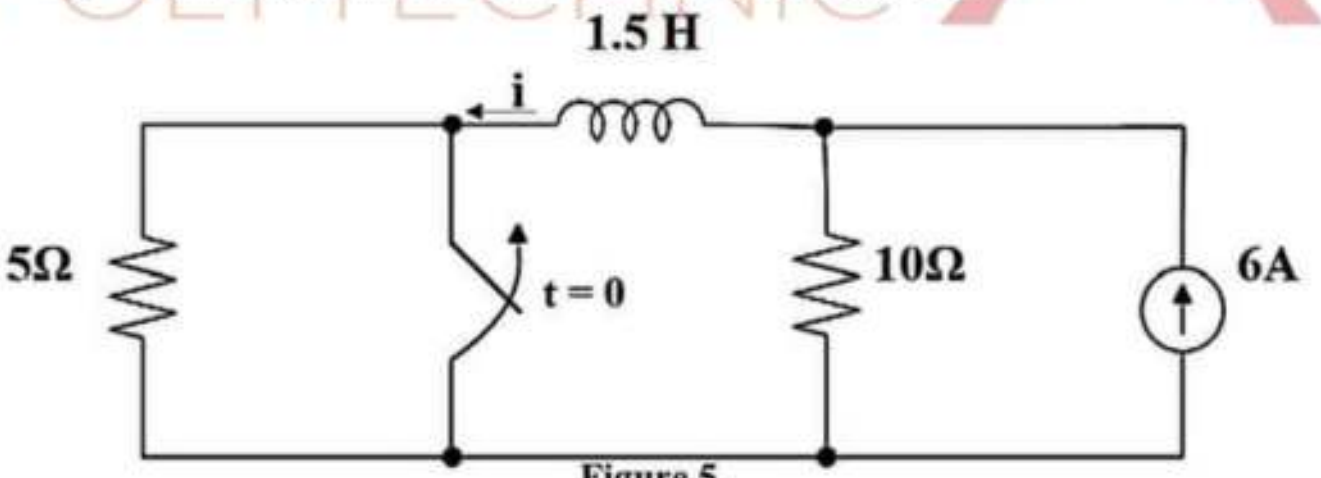
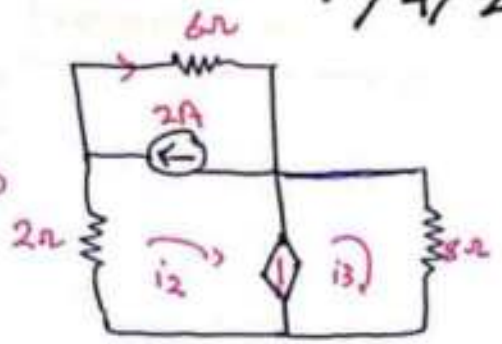


Figure 5.

Q1

1/7/2017

Mesh



$6i_1 + 8i_3 + 2i_2 = 0$ — ①

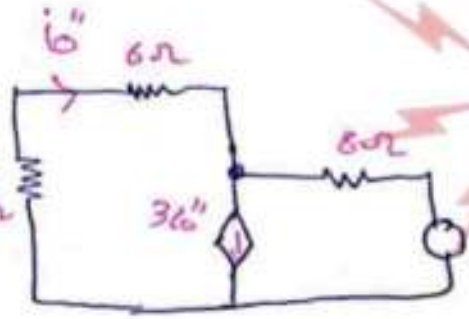
$i_1 - i_2 = 2$ — ②

$i_2 - i_3 = 3i_o'$ — ③

$i_o' = i_1$ — ④

$i_o' = -2.5\ A$

#spark-team
sultanabushahab



$\frac{U_1}{8} + \frac{U_1 - 24}{8} + 3i_o'' = 0$

$i_o'' = \frac{-U_1}{8}$

$U_1 = -24\ V$

$i_o'' = 3\ A$

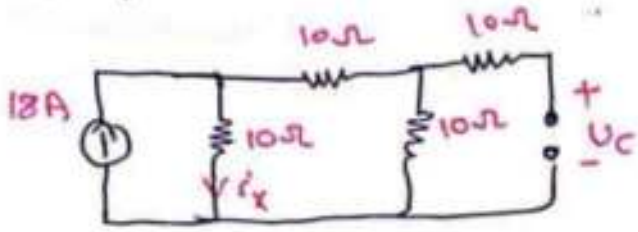
$i_o = i_o' + i_o'' = -2.5 + 3 = 0.5\ A$

$P_{6\Omega} = i^2R = 0.5 \times 6 = \frac{6}{4} = 1.5$ Watt

spark_team

Q3

using current divider



$i_x = 18 \times \frac{20}{10+20} = 12A$

$i = 18 - 12 = 6A$

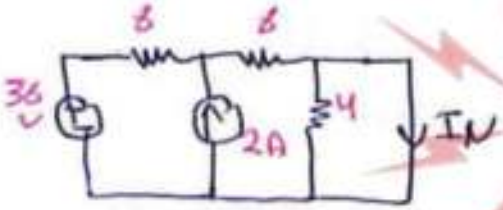
$V_c = 10i = 60V$

$W_c = \frac{1}{2} C V_c^2 = \frac{1}{2} \times \frac{5}{100} \times 3600 = 90J$

Q4

I_N ??

using supermesh



$6i_1 + 6i_2 = 36$

$i_1 + i_2 = 6$ --- (1)

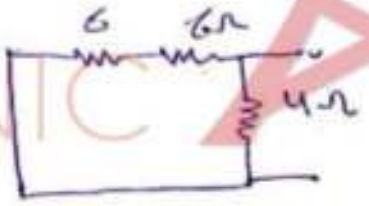
$i_2 - i_1 = 2A$ --- (2)

$i_1 = 2A \quad i_2 = 4A$



$I_N = i_2 = 4A$

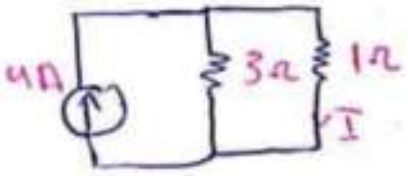
R_N ??



6 series 6 = 12Ω

12//4 = 3Ω = RN

$I = \frac{4 \times 3}{3+1} = 3A$



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sultan abusalah

Q5] $L = 5H$

$i(t) = 10 - 30t$ A

$v(0) = 0$ V

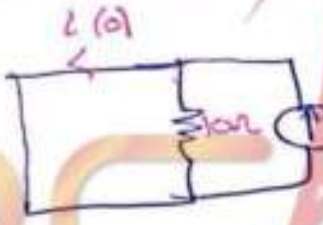
$$v_L = L\frac{di}{dt} = 5 * \frac{d(10-30t)}{dt} = 5*-30 = \boxed{-150V}$$

WL of t = 1

$$W_L = \frac{1}{2}Li^2 = \frac{1}{2} * 5 * (10-30(1))^2 = \frac{1}{2} * 5 * 400 = \boxed{1000\,J}$$

Q6] initial condition

$i(0^-) = i(0^+) = 6A$



$$i(\infty) = \frac{10}{5} = \boxed{4A}$$

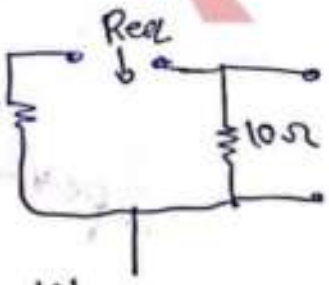


$i_f = I_0(1 - e^{-t/\tau})$ $i(0) =$

$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$

$i(t) = \left(4 - \frac{4}{2}\right)e^{+t/\tau} = 0$

$\tau = \frac{L}{Req}$



$i(t) = 6e + i_0 = 2e^{-t/\tau}$ $t \geq 0$

$i(200ms) = 2e^{-\frac{0.2}{\tau}}$

$Req = 5 + 10 = 15$

$\tau = \frac{1.5}{15} = 0.1$ $\quad \therefore i = 4 + 2e^{-10t}$ A

$i(0.2s) = 2e^{-\left[\frac{0.2}{0.1}\right]} = 2e^{-2} = \frac{2}{(2.71)^2}$ Amp (4)

$i'(200ms) = 4 + 2e^{-10(200ms)}$

$= 4 + 2e^{-2} = \boxed{4.27A}$

Q4 Find $w, \alpha, v(t)$

$v(0) = 0 \quad I(0) = 2$



$\alpha = \frac{1}{2Rc} = 12500$

$w_o = \frac{1}{\sqrt{Lc}} = 3162.28$

$\alpha > w_o$ (over damped)

$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$

$s_1 = -\alpha - \sqrt{\alpha^2 - w_o^2} = -24593.4$

$s_2 = -\alpha + \sqrt{\alpha^2 - w_o^2} = -406.6$

$v(0) = 0 = A_1 e^0 + A_2 e^0 \Rightarrow A_1 = -A_2 \quad -①$

$\frac{v}{200} + i + c\frac{dv}{dt} = 0$

$\frac{v(0)}{200} + i(0) + C\frac{dv(0)}{dt} = 0 \rightarrow \frac{dv(0)}{dt} = \frac{-2}{0.2} = -10M$

$\frac{dv(t)}{dt} = A_1 s_1 e^0 + A_2 s_2 e^0 = -10M \quad -②$

$A_1 = 413.45 \qquad A_2 = -413.45$

$v(t) = 413.45e^{-2459.4t} - 413.45e^{-406.4t}$

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جامعة البلقاء التطبيقية

الاسم: _____

Electrical circuits 1: **Final Exam** **Second semester** **2017-2018**

Date: May. 15, 2018

Exam Time: 100 minuets

Choose the correct answer.

*** For the questions 1,2,3,4 consider the circuit shown in figure 1 and answer:**

1) The value of V_1 is:

a) 11.6V b)22V

c) -22V d)7.4 V

2) The value of the current (i_o) is:

a) 2A b) 2.8A

c)1.4A d) -1.4A

3) The value of the voltage (V_2) is:

a) -4.8A b) 4A

c) -2.4V d) 8.4v

4) The power absorbed by 4 Ω resister is:

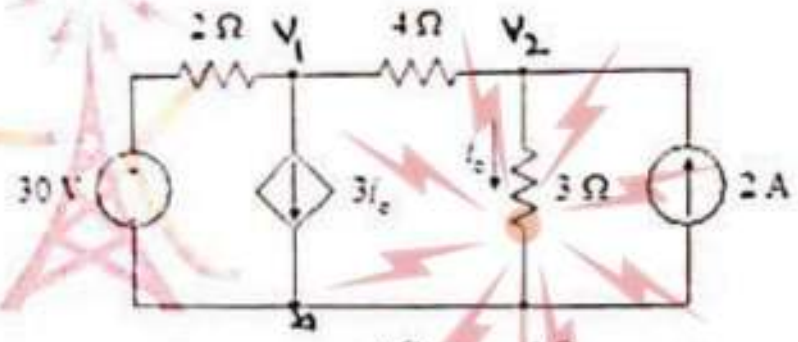
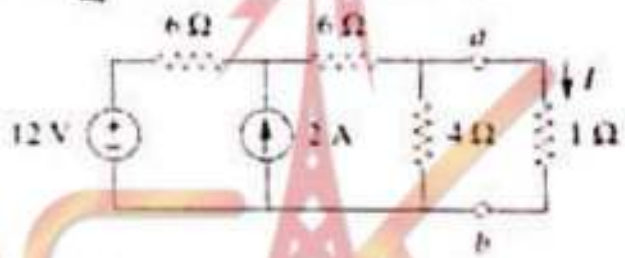


Fig.1

a) 6.4W b) 3.64W c) 2.56W d) 10W

*** For the questions 5,6,7consider the circuit shown in figure 2 :**



5) The value of Rth with respect to terminals ab is:

a) 3Ω b) 6Ω

c) 2Ω d) 17Ω

6) The value of V_{TH} from the terminals (a,b) is:

a) 3V b) 6V

c) 9V d) 12V

7) The power delivered to 1Ω resister is:

a) 2.25W b)1.5W c) 4W d) 4.45W

Fig.2

*** For the questions 8,9,10,11 consider the circuit shown in figure 3 and answer:**

8) The value of $v(t=\infty)$ is:

a) 15V b) -15V

c) 20V d) -20V

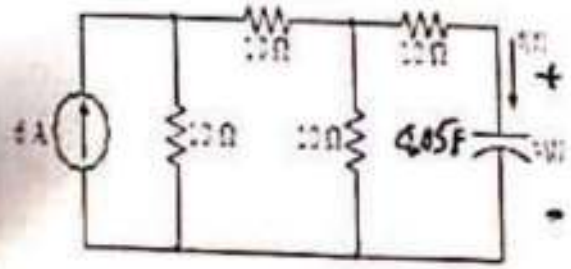
9) The value of i ($t=\infty$) is:

a) 2A b) 10A

c) 4.68A d) 0A

10) The value of r (time constant) is:

a) 1.2s b) 833 ms



c) 1.6 s d) 8.6 s

Fig.3

11)If The initial value of the capacitor voltage is v(0) = 10 V. The initial value of the capacitor current *i*(0) is:

a) 0.6A b) 1.2A c)10.63A d) 3.6A

*If the voltage across a 1 -mF capacitor is v(t) = 60 cos 100t mA, answer 12,13:

12) The current through the capacitor is:

a) -6 sin 100t mA. b) 6 cos 100t mA

c) -12 sin 100t mA d) -6 cos 100t mA

13) The energy stored is:

a) 1.8 $\sin^2(100t)$ μJ. b) 1.8 $\cos^2(100t)$ μJ.

c) 1800 $\sin^2(100t)$ mJ. d) 1800 $\cos^2(100t)$ mJ.

* For the questions 14,15,16,17 consider the circuit shown in figure 3 ($i(t) = 0.6e^{-2t}$ A. If i(0)= 1.4A) :



Fig.4

14) The current $i_{2(0)}$ is:

a) 0.8 A b) 1.6A

c) 6 A d) 3.2 A

15) The total current i2(t) is:

a) ($-0.8+1.4e^{-2t}$) A b) ($-0.4+1.2e^{-2t}$) A

c) ($-0.4+1.2e^{-2t}$) A d) ($-0.8+1.2e^{-2t}$) A

16) The voltage $v_1(t)$ is:

a) ($6+1.6e^{-2t}$) V b) ($18e^{-2t}$) V

c) ($6+1.6e^{-2t}$) V d) $-36e^{-2t}$ V

17) The voltage $v_2(t)$ is:

a) ($6-1.6e^{-2t}$) V b) ($6+1.6e^{-2t}$) V

c) $-7.2e^{-2t}$ V d) ($8+1.6e^{-2t}$) V

*For the questions 18,19 Refer to figure5:

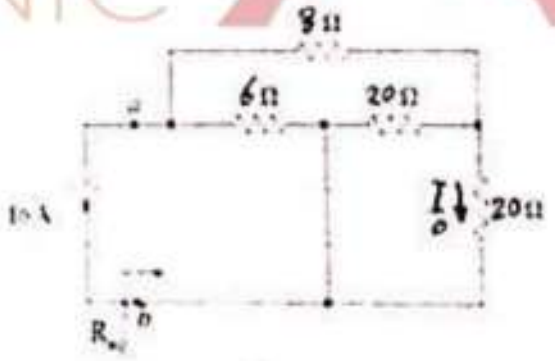


Fig.5

18) the value of the equivalent resistance is:

a) 9Ω b) 4.5Ω c) 6Ω d) 4Ω.

19) the value of the current i_o is:

a) 2A b) 4A c) 4.5A d)8A

• For the questions 20,21 consider the circuit shown in figure 6 :

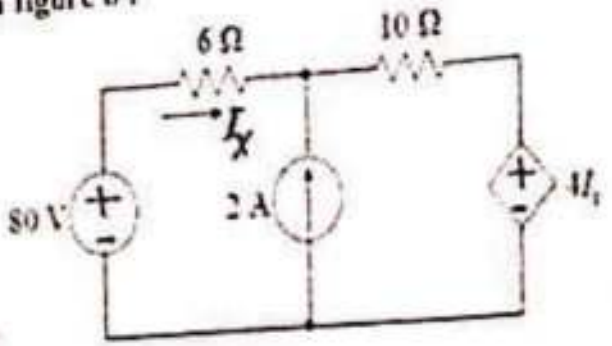


Fig.6

20) The voltage across 10 Ω resistor is:

a) 4 V b) 50 V
c) 90 d) 45 V

21) The current Ix is:

a) 5 A b) 2.4 A
c) -2.4 A d) 3 A

• For the questions 22,23,24 (consider the circuit shown in figure 7. Assuming $v(0) = 5V$, $i(0) = 0$, $L = 1H$, $C = 10mF$. And $R = 1.923\Omega$)



Fig.7

22) The value of α (damping factor) is:

a) 13 s^{-1} b) 52 s^{-1}
c) 26 s^{-1} d) 54 s^{-1}

23) The value of ω_o(resonant frequency) is:

a) 52 s^{-1} b) 10 s^{-1}
c) 20s^{-1} d)26 s^{-1}

24) The general natural response v(t) for t> 0 is:

a) $v(t) = A_1e^{s1t} + A_2e^{s2t}$ b) $v(t) = e^{-\alpha t}(A_1t + A_2)$
c) $v(t) = e^{-\alpha t}(A_1e^{j\omega_d t} + A_2e^{-j\omega_d t})$ d) $v(t) = e^{-\alpha t}(B_1\cos\omega_d t + B_2\sin\omega_d t)$

25) A heater draws 10 A when connected to 240V supply. How long does it take to consume 480 kJ.

a) 260 ms b)5m s
c) 20s d) 200 s

1	2	3	4	5	6	7	8	9	10	11	12	13
14	15	16	17	18	19	20	21	22	23	24	25	

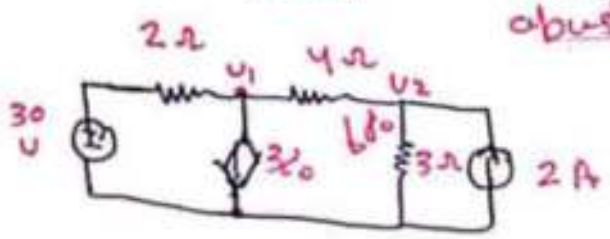
Good Luck

فايز برزين !

15/3/2018

sultan abusharab

① 11.6 V (A)



② $i_o = \frac{8.4}{3} = 2.8\,A$ (B)

③ 8.4 (D)

④ $P = \frac{v^2}{R} = v \cdot i = i^2 R$

$= \left(\frac{|11.6 - 8.4|}{4}\right)^2 \cdot 4 = 2.56\,W$ (C)

$\frac{v_1 - 30}{2} + 36 + \frac{v_1 - v_2}{4} = 0$

$i_o = \frac{v_2}{3}$

$2v_1 - 60 + 4v_2 + v_1 - v_2 = 0$

$3v_1 + 3v_2 = 50$ — ①

$\frac{v_2 - v_1}{4} + \frac{v_2}{3} = 2$

$3v_2 - 3v_1 + 4v_2 = 24$

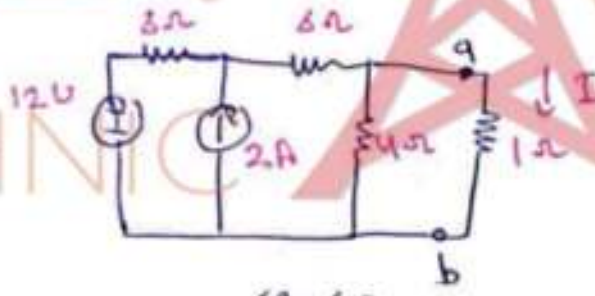
$-3v_1 + 7v_2 = 24$ — ②

$v_1 = 11.6\,V$

$v_2 = 6.4\,V$

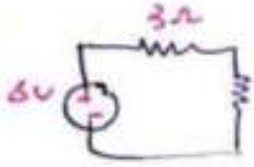
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⑤ 3Ω (A)



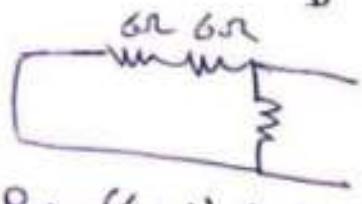
⑥ 6V (B)

⑦



$v_1 = \frac{1+6}{4} = 1.5\,V$

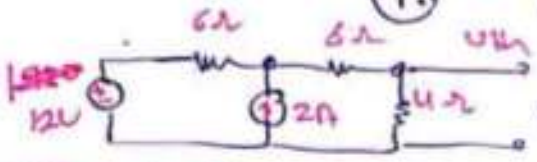
$P = \frac{v^2}{R} = \frac{(1.5)^2}{1} = 2.25\,W$ (A)



$R_{th} = (6+6) \,//\, 4$

$= 3\,\Omega$ (A)

(1)



$2 = \frac{v - 12}{6} + \frac{v - v_{th}}{6}$

$2v - v_{th} = 24$

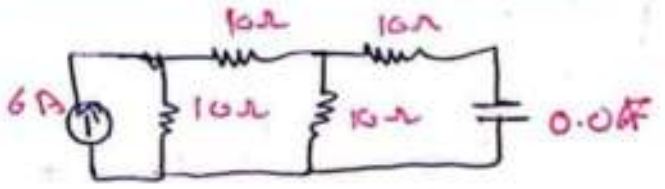
$\frac{v_{th}}{4} + \frac{v_{th} - v}{6} = 0$

$6v_{th} + 4v_{th} - 4v = 0$

$-4v + 10v_{th} = 0$

$v_{th} = 6\,V$

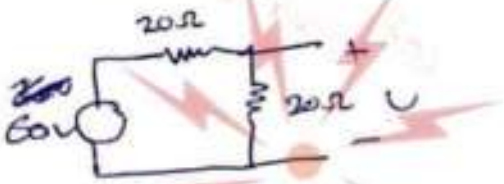
(8) 20 V (C)



(9) Zero A (E) (D)

(10) 833 ms (B)





(11) $v(0) = 10\ V$

$v(t) = 20 + [10-20]e^{-t/0.5}\cos$

$= 20 - 10e^{-12t}$

$i = C\frac{dv}{dt}$

$i = 0.05 \times 12\, e^{-12t}$

$i = 0.6\, e^{-12t}$

$i(0) = 0.6\ A$ (A)

$V = \frac{10 \times 60}{30} = 20$ volt



$R_{th} = (10+10) \| 10) + 10$

$= 16.66\ \Omega$

$\tau = 0.05 \times 16.66$

$= 833$ ms

(12) $C = 1\ MF$ $v(A) = 6\cos 100t$ mA

$i = C\frac{dv}{dt}$

$= 0.01 \times (-6000 \sin 100t)$ (A)

$= -6 \sin 100t$ mA

(13) $w = \frac{1}{2} C v^2$

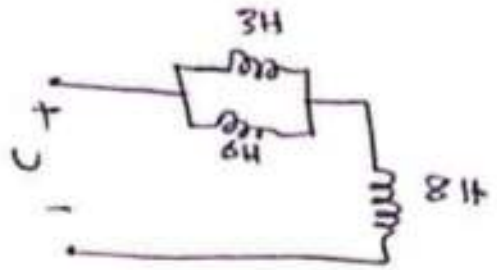
$= \frac{1}{2} \times 0.01 \times (60\cos 100t \times 10^{-3})^2$

$= 1.8\cos^2 100t$ mJ (B)

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(14) $i(t) = 0.6e^{-2t}$

$i(0) = 1.4\ A$



$i(0) = i(A) + i(B)$

$1.4 = 0.6 + i_2(0)$

$i_2(0) = 0.8\ A$ (A)

(15) $U_{6H} = U_{3H}$

$U = L\frac{di}{dt} = 6 + 0.6 \times -2e^{-2t}$

$= -7.2e^{-2t}$

$i_2(t) = \frac{1}{L}\int_0^t U\,dt + i_2(0)$

$= \frac{1}{3}\int_0^t -7.2e^{-2t} + 0.8$

$= \frac{1}{3}\left[\frac{-7.2}{-2}e^{-2t}\right]_0^t + 0.8$

$= 1.2e^{-2t} - 1.2 + 0.8$

$= 1.2e^{-2t} - 0.4$ (B)

(16) $i(t) = i_1(t) + i_2(t)$

$= 0.6e^{-2t} + 1.2e^{-2t} - 0.4$

$= 1.8e^{-2t} - 0.4$

Req

$U = L\frac{di}{dt} = 10 + 1.6 \times -2e^{-2t}$

$= -36e^{-2t}$ (D)

(17) $U_{6H} = U_{8H}$

$U = L\frac{di}{dt} = 6 + 0.6 \times -2e^{-2t}$

$= -7.2e^{-2t}$

(C)

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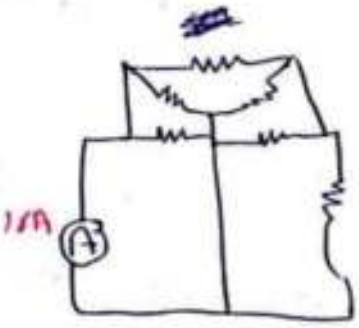
(18) $R_{eq} = ((20+47)//3.53) + 1.411$

$R_{eq} = 4.5\,\Omega$ (B)

$3.0886\,\Omega + 1.411 = 4.5\,\Omega$

$20 + 4.7 = 24.7$

$24.7 // 5.53 \rightarrow 3.0886$

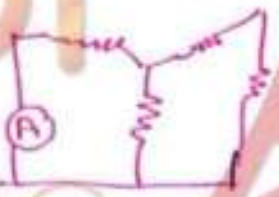




$R_1 = \frac{6+8}{34} = 1.411\,\Omega$

$R_2 = \frac{20+6}{34} = 4.7\,\Omega$

$R_3 = \frac{30+6}{34} = 3.53\,\Omega$



(19) $U_{3.0886} = 3.0886 + 16$

$= 49.47736$

$I_{24.7} = \frac{49.4736}{24.7} = 2\,A$ (A)

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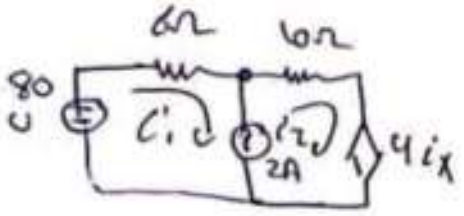
(20) $i_x = 4$

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$-80 + 6i_1 + 10i_2 + 4i_x = 0$

$6i_1 + 10i_2 + 4i_1 = 80$

$10i_1 + 10i_1 = 80$ ... ①



$i_1 + 2 = i_2$

$i_1 - i_2 = -2$ ... ②

$i_1 = 3A$

$i_2 = 5A$

$i_x = 3A$

(20) $V_{20} = 5 \times 10 = 50\ V$ (B)

(21) $i_x = 3\ A$ (D)

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(22) $v(0) = 5$ volt $L = 7H$

$i(0) = 0$ $C = 10\mu F$

$R = 1.923\Omega$

$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 1.923 \times \frac{10}{0.01}} = 2.6\ s^{-1}$ (C)

(23) $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 0.01}} = 10\ s^{-1}$ (B)

(24) $\alpha > \omega_0$ $v(t) = A_1 e^{-\alpha t} + A_2 e^{+2t}$ (A)

(25) $i = 10\ A$ $v = 240$ volt $w = 480\ kJ$

$p = i \times v = 10 \times 240 = 2400\ W$

$w = p \times t = 480 \times 10^3 = t \times 2400$ $t = 200\ s$ (D)

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